

Energy-Efficient Cooperative Spectrum Sensing: A Survey

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Abstract—The article analyzes the problem of energy efficient techniques in cooperative spectrum sensing (CSS). Although it was proven that single-device sensing is not sufficient for reliable sensing, cooperative spectrum sensing was proposed, burdened, however, with great overhead. Thus, work on the topic of energy efficient cooperative schemes gained more interest, which resulted in a number of energy efficient cooperative algorithm proposals. In this work, we try to classify the possible directions in energy efficient CSS and present a limited set of works introducing new ideas to an energy efficient CSS algorithm.

Index Terms—Cognitive radio, cooperative communication, energy efficiency.

I. INTRODUCTION

IN RECENT years, an exponential growth of the mobile data-traffic volume has been observed in radio communication systems [1]. The ever-increasing demand for higher data rates in these systems calls for new methods of handling and managing radio frequency resources. These resources are scarce. On the one hand, there are few available frequency bands that could be assigned to new radio communication systems, while on the other, a number of measurement campaigns conducted in many places world-wide have shown that a lot of frequency bands licensed to such systems are significantly underutilised [2]–[13]. This observation has inspired researchers to come up with the idea of cognitive radio (CR), introduced in fundamental works by J. Mitola III [14]–[16]. These publications have also stimulated investigation in the field of efficient spectrum sharing and dynamic spectrum access; new visions on the CR systems have been also proposed, e.g., in [17], [18].

Consequently, CR technology has been proposed as a potential solution to increase the efficiency of spectrum utilisation, as it enables opportunistic access of temporarily unused frequency bands once the presence of so-called Primary Users (PUs), i.e., the users of licensed systems, is excluded. The main point of this idea is to determine, with possibly the highest probability, whether the considered frequency band is available, i.e., not occupied by a PU. This may essentially be achieved in two ways: *i*) based on the knowledge about scheduled primary activities (e.g., stored in a database), or

ii) based on the real-time measurement of the PU's activity, known as spectrum sensing. The latter solution was initially considered as the main one for future CR systems, but due to long-standing open research issues in the implementation of reliable sensing methods, the interest in databases grew significantly [19], [20]. However, investigation on spectrum sensing is still highly encouraged, as sensing can complement and extend the information provided by databases.

This may be observed in recent standardisation activities. For example, the IEEE 1900.6 working group has started a new standardisation project on the topic of the usage of sensing information to support spectrum databases assigned as IEEE 1900.6b [21]. This is a third part of standardisation efforts started by publishing the baseline standard IEEE 1900.6 in 2011 [22] related to spectrum sensing interfaces, continued by the release of the first amendment (assigned as 1900.6a) in June 2014, entitled: "Procedures, Protocols and Data Archive Enhanced Interfaces" [23]. Moreover, spectrum sensing is a function of the *cognitive plane* in the IEEE 802.22 standard of Wireless Regional Area Networks (WRAN) in TV bands [24], [25]. In order to protect the Primary Users (e.g., television receivers and Programme Making Special Event (PMSE) equipment), apart from using the spectrum databases, spectrum sensing is compulsory for base stations and Consumer Premise Equipment (CPE) using this standard. However, IEEE 802.22 does not recommend a specific sensing method, since it only defines the detection level, SNR, time and detection quality requirements.

The next motivation for a continued research on spectrum sensing is the creation of the so-called Radio Environment Maps (REM), or other databases designed for storing and processing of the reach available context information. The justification for practical deployment of such databases relies on the assumption that the more (true and verified) information available in the system, the better its efficiency. The problem of accurate channel state information at the transmitter and receiver in a classical example proves this somehow trivial observation. However, the concept of utilising of the so-called context information can be easily extended to other aspects of wireless communications. *Information crumbles* collected by the system at a certain time and location can be smartly utilised in the future. In the context of 5G, or in the future, next-generation systems, the information about spectrum utilisation (including some historical knowledge and past experience, etc.) will be the basis of flexible spectrum management foreseen as one of the key technological enablers, even if the ideal cognitive radio concept is not applied in practice. The deployment of REMs and their advanced management is a practical tool

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for achieving this goal. In that context, cooperative spectrum sensing, decision making and reporting will then be used as an efficient way of permanent spectrum monitoring and delivering periodic updates of database entries. Thus, it is necessary to define energy-efficient solutions for cooperative spectrum sensing. Further reading on REMs and their applications can be found in, e.g., [26]–[31].

Besides the effective spectrum utilisation, the overall energy efficiency of a wireless network has been recognised as the key paradigm of the future 5th generation (5G) radio communication systems [32]. This is because the mentioned exponential increase of mobile data traffic significantly contributes to the world-wide consumed energy and related CO₂ footprint. For the future communication, Energy Efficiency (EE) of wireless systems is required to be improved by the factor of 10. In these systems, in which cognition capabilities are in place, the protection of PU transmission from the interference generated by secondary users (SUs) of the CR system is a prerequisite. As PUs can start and complete transmission at any moment, sensing should be the permanent process, and it may consume a considerable amount of energy. Thus, energy-efficient sensing is an important issue, and it has been addressed by a number of studies.

This survey article merges the mentioned spectrum and energy efficiency: on the one hand, it considers the idea of spectrum sensing, on the other the aspect of energy efficiency is underlined and relevantly discussed. The main goal of this paper is to analyse, and classify various spectrum sensing methods according to the possible ways of energy savings; the presented classification is also the main novelty of this contribution. Following the survey analyses, e.g., as presented in [33]–[38], where specific aspects of spectrum sensing are provided, and in [39], where the optimum spectrum sensing is discussed, in this work, we intentionally focus on energy efficiency in cooperatively-sensing networks.

The paper is organised as follows. First, the fundamental paradigms of cooperative spectrum sensing (CSS) are reviewed in Section II, followed by the discussion on reliable metrics used for energy efficiency assessment. Section IV provides a high-level classification of the possible areas of energy efficiency in CSS, whereas the consecutive sections deal with their detailed description. In particular, the optimisation of the single-node spectrum sensing procedure is presented in Section V, discussion on the required number of cooperating nodes is provided in Section VI, possible fusion schemes are considered in Section VII, while the analysis of energy-efficient network organisation solutions is given in Section VIII. Finally, eventual classification of energy saving approaches is presented in Section IX and conclusions are drawn in Section X.

II. COOPERATIVE SPECTRUM SENSING

The main goal of spectrum sensing is to identify the presence or absence of a PU at a certain location, at a given moment, and in a specified frequency band (Fig. 1). Spectrum sensing in its simplest non-cooperative form is considered as single-device (or single-node) sensing, where each node makes an independent decision on the availability of a frequency band, and acts accordingly (transmits in this band or not). From

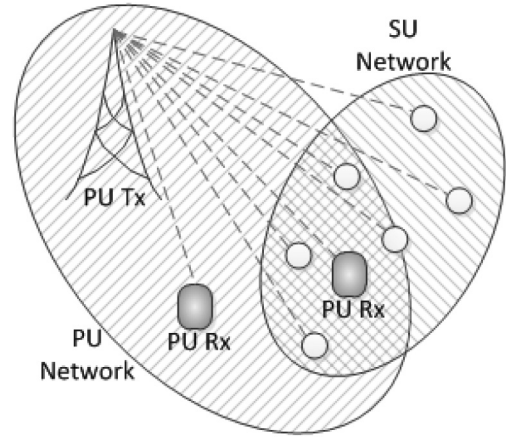


Fig. 1. Illustration of the primary and secondary user coexistence in the cognitive radio network.

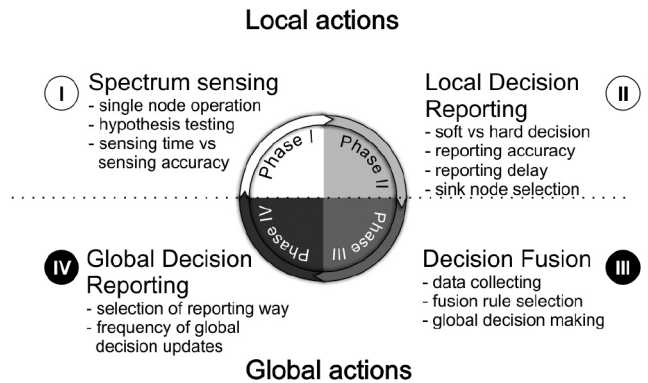


Fig. 2. Cooperative spectrum sensing procedure.

this perspective, numerous spectrum sensing algorithms have been proposed, such as the ones described in [34], [40]–[43]. However, several investigations pointed out that sensing carried out locally by single devices is not accurate enough for the safe coexistence of primary and secondary users [44]–[46]. Thus, it is generally agreed that one of the ways to increase the reliability of spectrum sensing is to apply cooperation between nodes. In cooperative spectrum sensing every node in a cognitive network senses the spectrum, and reports local sensing results, which are then used for acquiring a global decision characterised by the global probability of detection (see Fig. 2). In the following subsections each of these phases will be discussed in detail.

A. Single-Node Spectrum Sensing

The goal of spectrum sensing is for the cognitive entities (secondary users – SU) to decide on the presence of a licensed-system signal (a PU signal). If a SU detects signal $r(x)$, the spectrum sensing decision $D(x)$ can be treated as a double-hypothesis statistic test:

$$D(x) = \begin{cases} \mathcal{H}_0 & \text{if } r(x) = n(x) \\ \mathcal{H}_1 & \text{if } r(x) = s(x) + n(x), \end{cases} \quad (1)$$

where $\mathcal{H}_0$ is the hypothesis that the received signal is just noise $n(x)$, i.e., the frequency band is vacant, and $\mathcal{H}_1$ is the hypothesis that the $r(x)$ is the sum of noise and PU signal $s(x)$, i.e.,

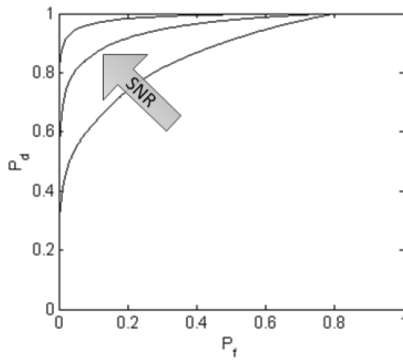


Fig. 3. Receiver operating curves for detection of frequency modulated signal.

spectrum is occupied. The sensing quality is described by the probability of detection P_d , i.e., the probability that $D(x) = \mathcal{H}_1$ in the case when the PU is, in fact, active, and by the probability of miss-detection P_{md} , i.e., the probability that $D(x) = \mathcal{H}_0$ in the same case, when the PU is active. Moreover, an important sensing-quality metric is the probability of false alarm P_f , i.e., the probability that $D(x) = \mathcal{H}_1$ in the case when the PU is actually not active. In general, it is beneficial to maximise the detection probability and minimise the false alarm probability. However, the optimisations of these two metrics are contradictory goals, i.e., an increase of the detection probability (by lowering the requirements for the decision threshold) leads to a higher number of false alarms and consequently to an increase of the probability of false alarm. Similarly, raising the detection threshold lowers both the detection probability and the probability of false alarm. Thus, for every sensing method, a trade-off has to be found that aims at keeping a possibly low number false alarms while guaranteeing a high detection rate. This constituted trade-off is shown in Fig. 3, where a few so-called receiver operating curves (ROCs), presenting P_d vs P_f , have been illustrated for various signal-to-noise ratios (SNR). One can observe that the higher the SNR, the better the receiver characteristic, understood as a lower false alarm rate for a given detection rate or a higher detection rate for the stated false alarm rate.

Note that there exist a number of single-node sensing methods, which may be blind or make use of some a priori knowledge of noise- and detected signal characteristics. Some recent results regarding various sensing methods may be found in [47]–[52], while surveys of these sensing techniques can be found in [34], [35], [53]–[55]. After every node acquires sensing information with a given quality measured by P_d and P_f , the next step is to share this information with other sensing entities in order to come up with a global, improved-quality decision.

Identified energy versus quality trade-off in single-node sensing In the context of energy saving, it is easy to indicate the following trade-off between sensing accuracy and certainty on the one hand, and processing time on the other. In a nutshell, the longer the sensing time or the more complex the sensing procedure, the more accurate the decision made by a single node (the lower the probability of interference that could be generated by a SU to a PU system and the better the frequency band utilisation). However, a more accurate sensing procedure requires the

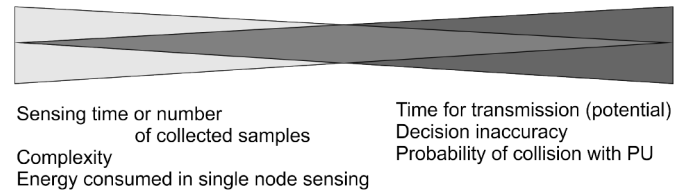


Fig. 4. Energy-efficiency trade-off observed in single node sensing phase.

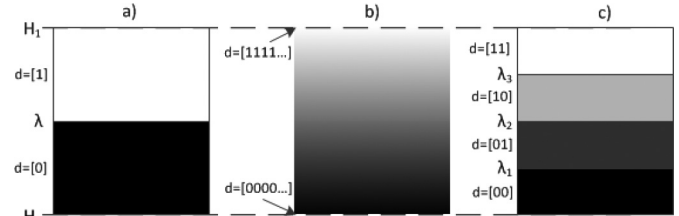


Fig. 5. Spectrum sensing possible types of reported decisions: a) hard-decision, b) soft-decision, c) double-bit hard decision.

shortening of the transmission phase and an increase of energy consumption in the sensing phase. This trade-off is graphically illustrated in Fig. 4.

B. Local Sensing Information Reporting

One of the simplest techniques for sharing the sensing information is based on the nodes' local decisions on the PU presence in a given band and on delivering local binary decisions to selected node(s) determined by a MAC layer. This procedure is known as hard-decision combining and requires just one bit for the local decision representation (e.g., 1 can represent spectrum occupancy, whereas 0 – vacancy). It is very concise and easy-to-decode, therefore, decision fusion may be easily adopted here (see Sec. II-D). However, such a concise quantisation of sensing information may be imprecise, and the quality of the decision made by the node is not reflected in its binary representation. In Fig. 5a, one may observe that sensing information is binarized, i.e., only presence/absence messages are generated. If the value of the decision variable (such as the received power, the presence of periodicity in the received signal, etc.) is above a specified threshold (denoted here arbitrarily as λ), the node will decide on the hypothesis $\mathcal{H}_1$ and the message will be $d = [1]$, otherwise it will be $\mathcal{H}_0$ with $d = [0]$.

In the soft-decision reporting scheme, soft sensing-information with an assumed level of accuracy is encoded and shared. In this scheme, the reported message may be extended by additional useful information, such as sensing-channel quality information, sensing decision quality, a dedicated metric characterising the node's previous decisions etc. Thus, soft-decision reporting may be very precise and neatly used in order to increase global detection quality; for example, one may find an optimum soft-decision scheme based on the Neyman-Pearson criterion in [56]. However, soft-metric reporting is burdened with large data overhead and computational complexity due to the large size of reporting messages. Additionally, large reporting messages introduce additional delay in transmission. From the perspective of energy efficiency, the size of

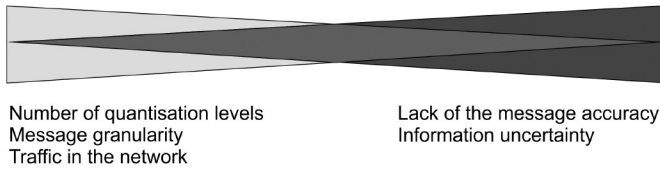


Fig. 6. Energy-efficiency trade-off observed in single node reporting.

reported messages should be limited: on one hand, precise soft reporting information leads to a high detection quality but also a large size of messages; on the other hand, limited precision provides a lower detection rate but also reduces the overhead significantly. Such a situation is illustrated in Fig. 5b, where no decision threshold λ is identified. Message d shared by the node can be used to somehow reflect the degree of uncertainty of the node's decision, e.g., the vector of ones $d = [111 \dots 1]$ will correspond to the certainty of the correctness of hypothesis $\mathcal{H}_1$. The higher the number of bits in vector d , the higher the accuracy of the decision. A comparison between hard-decision and soft-decision reporting is provided, e.g., in [57], where in the latter scheme, all nodes transmit information on the observed signal energy to the fusion centre.

However, in practical applications, the soft decisions have to be quantised with an assumed and rather low number of bits. This leads to solutions known as *quantised-soft reporting* which merge soft and hard reporting. As an example, the so-called quantised soft-bit schemes were proposed, and discussed in [58], [59]. In particular, in double-bit reporting, two bits are used in reporting messages, which gives four cases in reporting [56], as shown in Fig. 5c. A detailed analysis of the aforementioned schemes is presented in [60]–[62], whereas in [63], various quantisation schemes applied to reported message are analysed. Moreover, some discussion on the energy efficiency of various reporting and fusion rules is presented in [64].

Identified trade-off in sensing information reporting. Again, in the context of energy saving, it is easy to indicate the following trade-off between the accuracy and granularity of transmitted information and the energy and time needed for this action. The higher the granularity of the transferred information, the higher the generated traffic in the network and energy consumption in that particular phase, but at the same time, the more detailed and accurate the message delivered to the fusion centre. This trade-off is graphically illustrated in Fig. 6.

C. Selection of the Sink Node

The preparation of sensing information that will be put into the reporting message is followed by the process of sharing (spreading to selected nodes) the local observations. There are several possible configurations for spreading these local observations to other nodes. This essentially relies on local conditions such as: the presence of a fusion centre (FC), the qualities of sensing channels or the availability of a reporting channel [62].

One of the simplest possibilities is to select the central entity, namely the FC, and send (report) the local sensing information to this entity. The FC collects sensing-decisions from the nodes, generates a global decision, and broadcasts it back to nodes.

In Fig. 7a, such a centralised topology is presented. The black node which is the fusion centre is surrounded by several cognitive nodes reporting the sensing observations. The centralised topology is highly dependent on the proper FC selection: if the FC location is shadowed (in the sense of radio signal shadowing) the gain of such a centralised cooperative scheme is limited. Moreover, if the security aspects are crucial, the network sensing-performance should not be solely dependent on one node (the FC). Finally, the centralised scheme may have to deal with relatively large distances of some of the reporting channels, which may be inefficient energy-wise.

In the cluster-based scheme, the reporting channels distances are smaller. Geographically neighbouring nodes cooperate with each other, forming a closed group (called a cluster) with a selected cluster-head. All nodes report their local decisions to the local cluster head (instead of reporting them to the global FC), and it is the role of these cluster heads to either forward the collected decisions to the FC, or to make a decision at the cluster level, and report it to the FC. Such a case is illustrated in Fig. 7b, where the black node is the FC and the grey nodes are the cluster heads. The cluster-based scheme introduces a delay in sensing message sharing, and involves a specific procedure for organising nodes into clusters and the selection of cluster heads.

Unlike the centralised and cluster-based schemes, in the distributed topology, there is no selected single node that manages the sensing process in the network. Nodes interchange their observations in a somehow determined order, and make global decisions by combining their own observations with the ones acquired from messages sent by other nodes. The adopted distributed algorithm is determined by the applied distributed-network protocol. Distributed cooperative sensing is illustrated in Fig. 7c. There, every node takes a global decision on its own after collecting the sensing information from sensing entities. Note that the distributed sensing scheme may be slowed down by large number of signalling messages exchanged in the network.

Cooperation in a network may be implemented not only in the centralised, cluster-based or distributed manner. Relaying topology may also be adopted. Here, local sensing information is directed according to channel qualities experienced by a node. If its sensing or reporting channel is weak, the node may decide to cooperate with its neighbours in order to increase the detection probability. This cooperation may include relaying the reported sensing messages by another node to the FC or a selected node (if the direct reporting channel has poor quality). This is illustrated in Fig. 7d, where some channels between nodes are active, while others are not used. Note that the relaying procedure introduces the problem of delays in the network where the message is relayed by a number of nodes. Moreover, a high complexity of network management may be observed here.

Detailed discussions on sensing network topologies, efficient data exchange, clusters and cluster head management can be found in the rich literature on wireless sensor networks, such as [65]–[68].

Identified trade-off in the sink node selection. The trade-off that exists in the FC (also called sink-node) selection phase

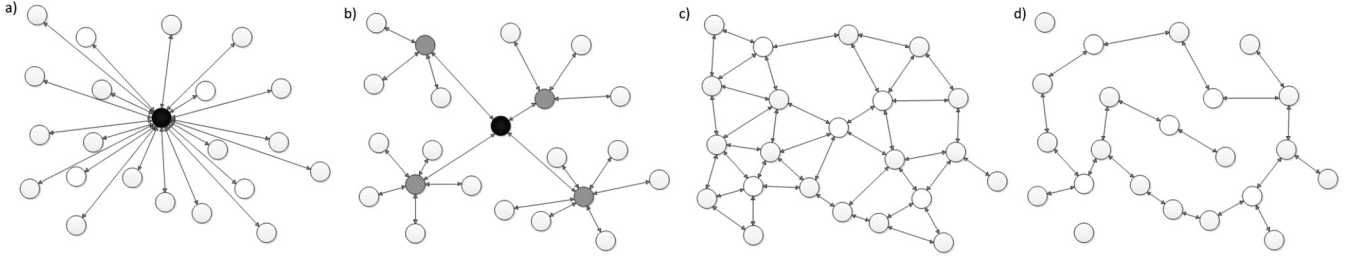


Fig. 7. Classification of cooperative sensing: a) centralised, b) cluster-based, c) decentralised and d) relayed.

TABLE I
PROS AND CONS FOR SELECTED TOPOLOGY

Topology	Pros	Cons	Reference
Centralised	- Detailed information acquired by the sink node, thus a more reliable decision made, - For N nodes, $N - 1$ messages have to be transmitted (fewer than in the decentralised case).	- Assuming that sensing messages are transmitted via a dedicated control channel, the need for delivery of $N - 1$ messages results in high quality-requirements for this channel, - The higher the number of nodes the longer the phase of single-node decision collection, - The more nodes, the longer the average distance between the sink node and the i-th node, thus, the more energy used for the transmission of sensing information.	[44], [69], [70]
Cluster-based	- The number of links is the same as for the centralised topology, - The time required for message delivery can be reduced by a factor close to the number of cluster heads, i.e., message delivery from individual nodes to cluster heads can be realised in parallel, - The average distance over which the message has to be sent is smaller compared to the centralised topology, thus lower transmit power can be used.	- Data processing in each cluster head may consume a considerable amount of energy, - In mobile networks, the frequency of invoking the cluster-head selection procedure may increase compared to the centralised-topology case, where the procedure of centralised node selection may be less frequent.	[39], [71]–[73]
Decentralised	- The number of links can be highly reduced, as each node makes a decision on its own, - If relaying is not allowed, each node collects sensing messages only from the surrounding nodes.	- If each node gathers information from all other nodes the number of sensing-messages transmissions grows significantly, - The energy required for data processing in each node may be high (the scheme may be inefficient in terms of energy consumption).	[74], [75]
With relays	- The number of links is similar as in the centralised-topology case, - Distances between neighbouring nodes are shorter than the average distance in the centralised scheme, - The time required for message exchange depends on the longest route in the sensing network, but is shorter than in the centralised-topology case.	- As in the cluster-based scheme, the energy required for data processing in each relay increases the energy consumption in a network, - There is a need for fast and accurate route selection, which also consumes energy and time.	[76]–[78]

can be characterised as follows. The more links between the nodes involved in exchanging of sensing information the longer the duration of this phase and the more energy is consumed for this purpose. However, each topology should be analysed separately, since the topology selection highly affects this trade-off. A simple comparison of pros and cons for each proposed topology is presented below in Table I.

D. Decision Fusion

After collecting the local sensing results, an effective fusion has to be performed by a selected sink-node (e.g., fusion centre). This is actually done by adopting the fusion rule [56], [79]–[81] and making the final decision [82]–[85]. Various rules are possible, and their comparison is possible after the adoption of relevant global metrics. These are associated with cooperative decisions based on local decisions acquired from

N secondary (sensing) nodes. Assuming the general k -out-of- N rule (where a positive decision is made only if the number of positive answers is not lower than k), the global probability of false alarm Q_f and the global probability of detection Q_d can be obtained as follows [86]:

$$Q_f = \sum_{i=k}^N \binom{N}{i} P_f^i (1 - P_f)^{N-i}, \quad (2)$$

$$Q_d = \sum_{i=k}^N \binom{N}{i} P_d^i (1 - P_d)^{N-i}, \quad (3)$$

where P_f and P_d are averaged over the statistics of N nodes, so for instance, P_f is equal to:

$$P_f = \frac{1}{N} \sum_{i=1}^N P_{f,i}, \quad (4)$$

and $P_{f,i}$ represents the probability of a false alarm of the i -th node. General formulas (2) and (3) may be modified in order to express three widely used fusion rules: the *OR*, the *AND* and the *majority* rule. Moreover, for any specific rule, the global probabilities of false alarm and detection are obtained in an analogous way, therefore we present only formulas related to detection quality. In the case of the *AND*-rule (N -out-of- N rule), and the *OR*-rule (known as 1-out-of- N rule) the above formula for Q_d is simplified to:

$$Q_d^{\text{AND}} = \prod_{i=1}^N P_d = P_d^N, \quad (5)$$

$$Q_d^{\text{OR}} = 1 - \prod_{i=1}^N (1 - P_d) = 1 - (1 - P_d)^N, \quad (6)$$

respectively. Furthermore, Q_d for the majority-rule, when at least half of the nodes have to detect the PU activity, is given by:

$$Q_d^{\text{MAJ}} = \sum_{i=\lceil N/2 \rceil}^N \binom{N}{i} P_d^i (1 - P_d)^{N-i}. \quad (7)$$

The Constant False Alarm Rate (CFAR) scheme assumes that in the network, the global probability of false alarm Q_f remains constant, i.e., is set for the whole secondary network. Thus, the corresponding value of $P_{f,i}$ is assumed to be identical for every node, and can be obtained for the OR rule as:

$$P_{f,i} = 1 - \sqrt[N]{1 - Q_f} \quad \text{for } i = 1 \dots N. \quad (8)$$

Thus, the aim of the CFAR scheme is to maximise the received probability of detection. On the other hand, under the Constant Detection Rate (CDR) requirement, the value of Q_d is set for the network, while the false alarm probability is minimised. The network starts in this case with assumed local probabilities of detection for each node (which are stated in an analogous way as for CFAR).

Identified quality versus energy-efficiency trade-off in the decision fusion. The analysis of the quality versus energy-efficiency trade-off, in the case of decision fusion, is not straight-forward. Let us focus on the OR rule. The application of this approach results in a high probability of detection and a high probability of a false alarm, thus also in a low probability of a potential collision with a PU. However, this also means that the probability of data transmission by a SU is reduced as well. Moreover, including the influence of the transmission channel, one can observe that the OR rule can result in the lack of identification of transmission opportunities. In such a case, the spectral efficiency (and thus also the energy efficiency) is very poor. On the other hand, the AND rule can be described in a *reversed* way. Thus, one can generally say that there is a trade-off between the certainty of the decision made by a fusion centre and the energy efficiency. However, this observation is not valid in all cases. If the probability of detection is low, the number of potential collisions with a PU increases leading to an increase of retransmissions for both the PU and the SU, and in consequence, to a lower energy efficiency. Finally, as mentioned above, typically, the assumed level of uncertainty (either

in terms of Q_d or Q_f) must be assumed as constant in the network. Thus, the selection of the rule will depend highly on the topology.

E. Global Decision Reporting

The last step in the CSS scheme is sharing (spreading) the global information and the global decision on the spectrum vacancy. In the centralised topology, until this step, every node is aware of the local decision. However, it is the awareness of the global decision that can create the expected gain in the cooperative network. To this end, the central entity spreads the global decision usually by sending a broadcast message. Then, based on this received information, the nodes plan future actions.

Unlike in the centralised scheme, in the distributed scheme, there is actually no need for sending the global decision. Here, the nodes acquire the information about their neighbours' observations during the local-decision reporting stage. Although this involves additional reporting messages, the decision fusion is performed by each node separately, and no broadcast message with a global decision is needed.

Finally, in the clustered scheme, the global decisions are sent back to the cluster-heads, and these nodes are in charge of the further distribution of this message within the controlled group of nodes.

Identified quality versus energy-efficiency trade-off in the global decision reporting. This phase of CSS is rather short, and requires the delivery of the final decision to all nodes. However, one of the questions that can be asked here is the following: is it more energy-efficient to broadcast the information to all nodes, or to reduce the transmit power and deliver this decision to the cluster heads, and allow them to further redistribute this message to more distant nodes?

F. Trade-Offs' Impact on Energy Efficiency

The identification of the existing trade-offs in each phase of CSS shows that there is a high number of degrees of freedom in the optimisation of the energy efficiency of cooperative sensing networks. Moreover, some of these trade-offs are mutually dependent, and overlap, making this problem even more challenging. In the following section, we try to structure this issue focusing on some selected key aspects of CSS.

III. CONSIDERATIONS ON ENERGY EFFICIENCY

As the spectral and energy efficiency have been identified as the key aspects of future wireless networks, it is important to decide on the accurate, reliable and effective way of energy efficiency measurement and assessment. This section provides a general overview of the energy efficiency aspects of CSS.

A. Time Domain Relations Between Spectrum Sensing and Spectrum Access

In general, SU needs to periodically sense the availability of the considered frequency band (either in a cooperative or

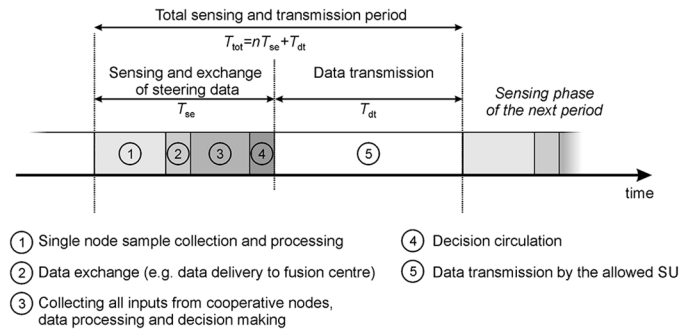


Fig. 8. Illustration of the sensing and data transmission period.

non-cooperative manner), and if the band is vacant, it can start or continue its transmission in this band (can access the spectrum). Such an exemplary situation is illustrated in Fig. 8, where one sensing and data-transmission period is shown. One can observe that the sensing phase consists of four sub-phases, i.e., the time needed for single-node spectrum sensing (including the collection of samples and their processing), the time required for decision delivery to the so-called fusion centre (in the centralised case) or its exchange among neighbouring nodes (in the distributed case), and finally, the time needed for data processing in the fusion centre and decision circulation among interested nodes. The total duration of this phase can be denoted as T_{se} . If the decision for a given SU is not positive, i.e., the considered frequency band is occupied, the spectrum sensing phase has to be repeated until the spectrum is free. Once it is available, the SU can transmit its data over period T_{dt} . Assuming the total number of spectrum sensing phases equals n , the duration of the sensing and data transmission period can be calculated as $T_{\text{tot}} = n \cdot T_{\text{se}} + T_{\text{dt}}$. This observation is true if the sensing node performs the sensing procedure asynchronously until it determines that the frequency band is vacant, i.e., the next portion of samples is collected either continuously or after a short period. The formula is different if the synchronous approach is applied, in which each sensing period T_{se} is succeeded by a transmission period T_{dt} , i.e., when the sensing node detects that the wanted spectrum is occupied, it waits over time T_{dt} until the next sensing period begins. Thus, for n tries, the total time equals $T_{\text{tot}} = n \cdot T_{\text{se}} + n \cdot T_{\text{dt}}$.

The above considerations lead to the conclusion that in order to maximise the energy efficiency of the sensing-based opportunistic spectrum access, the sensing period should be optimised to allow for reliable detection of spectrum opportunities, and at the same time, leave enough time for actual information-data transmission. This sensing-throughput trade-off has gained significant attention recently. On one hand, there exist a dozen of works treating the subject of sensing time optimisation (the most relevant ideas are described in Section V), on the other, there is a vast literature regarding the spectrum access. The exemplary works may be [87], where SU access is maximised through the penalties policy regarding interference with PU, and [88], where an adaptive random access protocol is described. However, the area of SU access, as well as medium access control (MAC) is beyond the scope of this article and will not be further analysed. For further reading

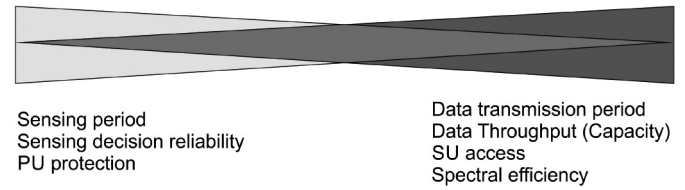


Fig. 9. Sensing period-data transmission period trade-off observed in spectrum sensing.

on spectrum access and MAC, we propose the comprehensive surveys [89], [90].

One of the key trade-offs in spectrum sensing is to adjust the sensing time and the spectrum-access time. The first one impacts sensing reliability (and thus, the PU protection and their throughput), the second – the SU spectral efficiency. Both impact the energy efficiency of a system: the longer the sensing time, the more energy is spent on sensing, the less time is left for actual SU transmission, the lower the throughput, and the lower the energy efficiency of the SU transmission. On the other hand, a longer sensing time increases the sensing reliability and decreases the interference level experienced by both the PUs and SUs, which results in higher PU links throughput and higher energy efficiency of their transmission. Therefore, the sensing period has to be carefully adjusted in order to guarantee a given level of sensing reliability (and PU protection), which is meant as a sufficiently low false alarm rate and an instantly high detection rate. A shorter sensing time and a longer data transmission period usually results in a higher spectral efficiency, due to higher throughputs and increased capacity. This trade-off is illustrated in Fig. 9.

As discussed above, optimisation of the sensing- and access-time alone might be contradictory goals, therefore, it has been proposed to analyse these two jointly. For instance, in [91], the authors suggest a joint optimisation of the spectrum sensing period and the transmission stage (spectrum access). They prove that there exists a suitable proportion between the sensing period and the access period guaranteeing the highest channel efficiency, i.e., a suitably low false alarm rate for a minimised sensing period. Similarly, in [92], under detection and energy-based detector constraints, an optimal sensing period is delivered that maximises the throughput. Secondly, cooperative detection is applied where the optimal sensing period is shortened, when compared to a non-cooperative scheme. This conclusion is reinforced in [93]. The cooperation adopted in [93], by forming coalitions, allows the improvement of sensing times and node link capacities. To this end, each SU may make an independent decision on joining or leaving the coalition based on a calculated utility, which jointly optimises the average sensing time and the average acquired capacity. Another joint optimisation of the sensing-and-throughput period is performed in [94]. There, SUs adopt a distributed learning algorithm in order to converge to an evolutionarily stable strategy, where some nodes sense the spectrum, while others may access it. The work puts emphasis on dealing with selfish users who want to access the spectrum without a contribution to sensing. By embedding rewards to the sensing contributors, in the learning algorithm, every node is inclined to apply the

required sensing strategy. Finally, in [95], the authors prove that the maximum throughput is gathered for a low number of reported bits and a moderate number of samples.

B. Identification of the Figure of Merits

Based on the above discussion, one can find several metrics related to power consumption in devices or networks. The basic one is the *bits-per-Joule* [b/J] which is essentially a throughput-per-energy usage metric. Throughput is understood here as the total number of bits transmitted in a network (so time T_{tot} is considered here). In the case of CSS, where a number of steering and control messages are often interchanged, the metric of *effective bit-per-Joule* can also be considered, where only the transmitted information-data bits are counted, and the bits transmitted in supporting channels are not included. Alternatively, the *bit-per second-per-Watt* metric can also be taken into account.

The next group of energy efficiency indicators is connected with network coverage, such as the *Watt-per-square kilometre* denoted as [W/km²]. This indicator is adequate for systems covering large areas, e.g., in macrocells used in rural areas or in sensor networks designed to cover a given area [96]. However, in urban areas, where a system is traffic-demanded, a more fair indicator is one related to the number of serviced users. Such a metric is defined as the number of users served in the busy hours measured in *users-per-Watt* [97]. An energy-related metric taxonomy may be found in [98]. Note that for cooperative sensing, these metrics have to be adapted to a specific CSS case. Contrary to cellular networks, in CSS, the coverage has to be understood as the area for which a decision made by the fusion centre is valid, or - alternatively - the area over which sensing nodes are deployed.

Although the coverage and capacity-related metrics are important and widely used, they are in fact not comprehensive enough in the cooperative spectrum sensing reality. We suggest to introduce energy efficiency metrics related to the *quality of sensing* understood as the *global probability of detection-per-Watt* and the *global probability of false alarm-per-Watt* [99]. Such metrics can be defined mathematically as $E_{Q_d} = \frac{Q_d}{P_{\text{tot}}}$, and $E_{Q_f} = \frac{1-Q_f}{P_{\text{tot}}}$, respectively. Here, Q_d and Q_f stand for the global probability of detection and the global probability of false alarm in the network, which have been defined in Section II-D. Moreover, P_{tot} represents the total power consumed by the network in one sensing and data-transmission period. The energy efficiency increases when the detection quality increases, and when the consumed power decreases, which are usually contradictory goals in the network.

C. Single Sensing-Node Power Optimization

Because energy efficiency may be measured with the use of a number of metrics, we focus our considerations on the reduction of the consumed power with as low a degradation of performance as possible. Thus, the first step in finding reliable solutions for energy reduction is to identify the possible opportunities where such optimisation can be achieved.

The power consumption in sensing node P_{node} consists of the energy devoted for sensing P_{sensing} , for the processing of gathered data and the preparation of the reporting message (e.g., quantisation) $P_{\text{processing}}$, and finally, for the transmission of this reporting message $P_{\text{transmission}}$

$$P_{\text{node}} = P_{\text{sensing}} + P_{\text{processing}} + P_{\text{transmission}} \quad (9)$$

Therefore, the optimisation of energy efficiency should include the reduction of the power related to each component from formula (9). In other words, the sensing energy may be reduced by shortening its duration, and the processing power usage depends on the complexity of adopted sensing method, as well as on the adopted quantisation algorithm applied for sensing-information representation (see Section II-B). The power consumed on the transmission of the reporting message depends then on the transmission distance and the environment in which the transmission is performed (described by the path loss coefficient). Besides, $P_{\text{transmission}}$ may be reduced by lowering the size of the transmitted message.

One may try to analyse, in a more detailed way, which factors influence the average power consumed by each sensing node in each phase. The amount of power consumed in the first phase, i.e., spectrum sensing P_{sensing} , depends on the electric and electronic components of the terminal front-end. In particular, in order to effectively sense a wide range of a frequency spectrum, the wireless terminal should be equipped - depending on the selected architecture (homodyne, heterodyne, low-IF etc. [100], [101]) with a wideband (possibly tunable) aerial, wideband and steerable filters and Low Noise Power amplifiers. Next, the received signal should also be shifted to the baseband, thus appropriate wideband mixers and voltage-controlled oscillators with phase-locked loops have to be used. Finally, such a signal has to be converted from the analogue to digital domain, and the power consumption by such tunable converters strongly depends on the processed bandwidth. Clearly, instead of wideband elements mentioned above, a set of parallel processing chains may be used. In order to estimate the power consumed by the analogue front-end of the sensing node, one should apply relevant models proposed in the rich literature, e.g., in [102], [103].

Let us now analyse the power consumption of the second phase. The power consumed during data processing $P_{\text{processing}}$ strongly depends on the computation complexity of the selected sensing algorithms. Such a computation complexity can be represented by the number of operations (e.g., complex multiplications, complex additions and the number of accesses to the memory) required for the preparation of the reporting message containing a local sensing decision. By getting the number of operations, one may assess the average power consumption by the DSP or FPGA modules (or other DSP components). Numerous discussions on the energy consumption by FPGA during digital processing can be found in the literature, e.g., in [104], [105]. A detailed comparison of selected single-node spectrum sensing algorithms (including computational complexity) can be found in [106].

Finally, $P_{\text{transmission}}$ represents the power required for reporting the message delivery to distant nodes (e.g., to the fusion centre). Depending on the situation (environment type, e.g.,

urban, suburban), appropriate channel models can be used for an accurate assessment of the power consumed by this data transmission.

Finally, let us highlight that, in fact, it is the energy, not power, which has to be considered during the optimisation process. This is due to the fact that the time devoted for each phase is also one of the parameters that can be optimised. For example, the selection of a specific sensing algorithm (such as energy detection) influences both the average power used for sensing, and the time required for it. Let us denote the sensing time as T_{sensing} , the processing time as $T_{\text{processing}}$ and the reporting time as $T_{\text{transmission}}$. The energy consumed by a single node in a CSS network equals:

$$E_{\text{node}} = P_{\text{sensing}} \cdot T_{\text{sensing}} + P_{\text{processing}} \cdot T_{\text{processing}} + P_{\text{transmission}} \cdot T_{\text{transmission}}.$$

D. EE Optimization From the Network Perspective

Each energy-efficient sensing node improves the overall energy efficiency in a cooperative network, but the total energy consumption may be further reduced if the energy consumption is analysed from a network-level point of view. In particular, the energy consumed in a centralised network of N cooperating nodes (in a network where a central entity collects sensing information and then announces a global decision by a broadcasting message) is given by:

$$P_{\text{network}} = \sum_{i=1}^N P_{\text{node},i} + P_{\text{fusion}} + P_{\text{broadcast}}, \quad (10)$$

where $P_{\text{node},i}$ is the power of sensing, processing and transmitting (reporting) the sensing information by node i in the network, P_{network} is the power devoted for CSS consumed in the whole network, P_{fusion} is the power devoted for the process of decision fusion, and $P_{\text{broadcast}}$ is the power for broadcasting the message by the fusion centre. Let us note that this formula can be easily adopted to other network topologies used for CSS. For example, in the approach known as censoring or node sleeping (described in Sec. VI), the number of nodes used for sensing and data reporting can be reduced. Moreover, the value of $P_{\text{broadcast}}$ depends mainly on the distance between the fusion centre and the most distant node awaiting its decision. In a centralised network, especially for large N , this component will be relatively small compared to the first one, i.e., the N -fold included power consumed by sensing nodes. Again, the time used for report delivery to the fusion centre and data broadcasting is also one of the parameters for optimisation. Thus, instead of the total power consumed in the network, we should concentrate on the energy consumed. The formula for energy consumption in a CSS network is the following:

$$E_{\text{network}} = \sum_{i=1}^N P_{\text{node},i} \cdot T_{\text{node},i} + P_{\text{fusion}} \cdot T_{\text{fusion}} + P_{\text{broadcast}} \cdot T_{\text{broadcast}}, \quad (11)$$

where $T_{\text{node},i}$ is the time required for sensing, processing and transmitting (reporting) the sensing information by node i

(assumed as a single activity time-period), T_{fusion} is the time required by the fusion centre for collected information processing and taking the global decision, and finally, $T_{\text{broadcast}}$ is the time of broadcasting the global sensing decision.

One can observe that the energy in the network may be saved by the reduction of the following factors: a) the number of cooperating nodes (e.g., by selecting only substantial nodes), b) the power consumed by the nodes (e.g., in the sensing, sensing-information processing or in the transmission phase for a selected subset of nodes), c) the energy cost of data fusion and decision delivery to the interested nodes (being a mixture of the two previous factors), d) or network topology.

To sum up, energy efficiency may be provided to the network with several possible methods. A variety of propositions which may be found in the literature, and are classified in this survey paper, show that the overall energy efficiency depends on the operating radio environment and allowed performance degradation.

E. Security

Network security is one of the key topics nowadays, and in particular, it is considered as an issue in the future cognitive radio networks. Most works discussed in this survey are based on the assumption that sensing results generated by nodes, as well as the fusion-centre decisions are not falsified by any source. However, without any countermeasures, the performance of CSS may be degraded, decisions falsified and appropriate protection of PUs threatened, which would make the whole idea of cognitive radio neither practical, nor implementable. Thus, the security aspects of cognitive radio have gained a lot of attention [40], [46], [107]–[109]. It is crucial to guarantee the reliability of the whole procedure of CSS. However, security measures come with an overhead of additional operations, complexity and extra energy consumption. One may figure out that the cost of security is in energy-efficiency reduction.

In [110], the impact of malicious users providing a false spectrum results is taken into account. The low-overhead protocol is proposed by finding the optimum number of extra bits (guaranteeing security) in terms of energy efficiency. In [111], another security-aided protocol is proposed. There, combined spectrum-sensing-data falsification (SSDF) and incumbent emulation (IE) attacks have been considered. The authors propose a procedure using lightweight cryptography in order to limit the energy-consumption overhead when protecting the CSS system against SSDF and IE attacks.

Another work [112] addresses energy efficiency indirectly. The authors focus on the fast detection of misbehaving SUs. The minimised time of detection of such a user reduces the energy consumption of the introduced security measures. In [113], new low-overhead methods are proposed, robust against the SSDF attacks, named the enhanced weighted sequential probability ratio test and the enhanced weighted sequential zero/one test.

IV. ENERGY EFFICIENCY IN CSS: CLASSIFICATION

The aforementioned analysis of cooperative spectrum sensing has set a solid background for the possible directions of

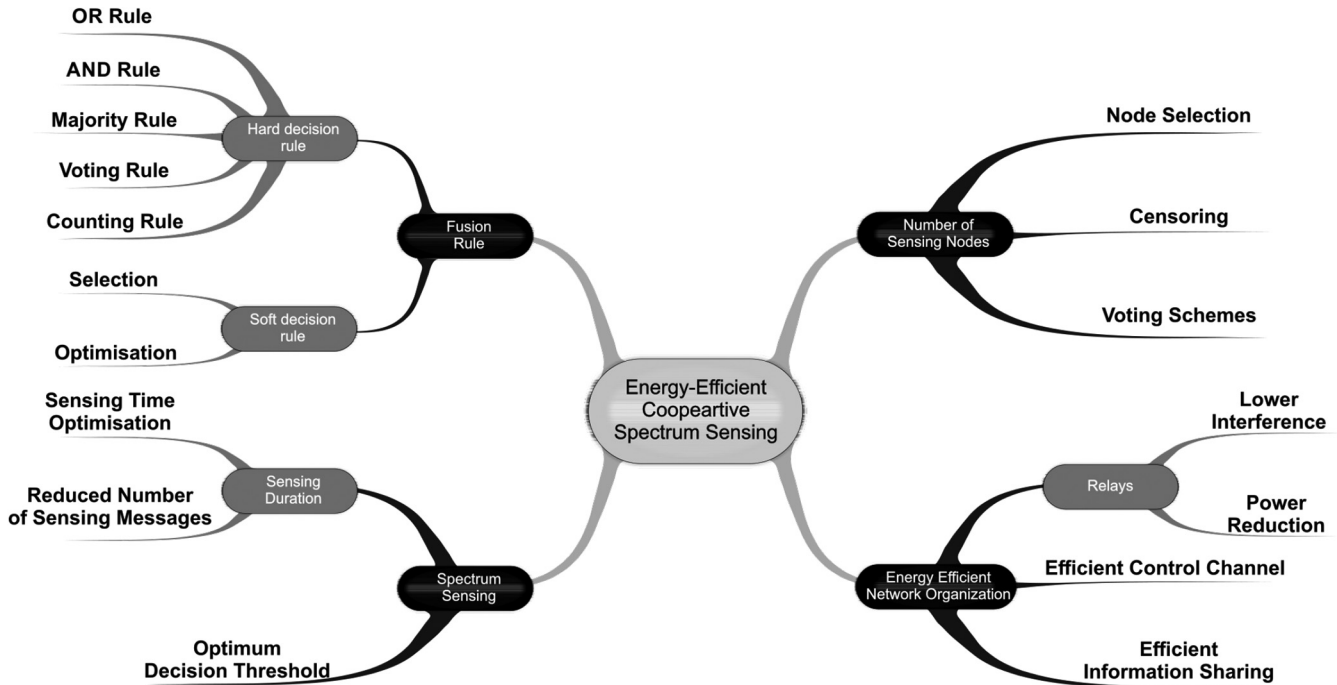


Fig. 10. Classification of energy-efficient cooperative spectrum sensing.

energy savings. In the literature, one can find the number of diverse energy-efficient techniques which may be grouped according to several possible classifications. In this work, we propose the classification based on four main directions (or branches) of possible energy savings (see Fig. 10). They are as follows:

- Branch A: energy reduction in the (local) spectrum sensing phase,
- Branch B: optimisation of the number of cooperating nodes,
- Branch C: proper selection and application of fusion and decision rules,
- Branch D: energy-efficient network organisation.

Below some general observations for each branch are provided, which will be discussed in detail in the following sections.

1) *Branch A*: Focusing on the first branch, one can state that an immediate and natural option of saving energy is to go inside the sensing device for the proper selection of energy-efficient components. The application of advanced and flexible radio-frequency front-ends, where some portions of digital or analogue elements can be switched off or shifted into standby mode may result in overall energy consumption reduction, especially if the number of sensing nodes is high. Exemplary discussions on such solutions may be found in [114]–[118], but these aspects will not be further analysed in this work.

Furthermore, one of the basic ideas of energy efficiency in local spectrum sensing is to reduce the sensing time or the number of collected samples regardless of the adopted sensing method. Of course, the lower the number of acquired samples, the lower the energy consumed. However, this comes at the cost of decreased performance. Solutions which optimise the sensing time (or the number of acquired samples) are

described in Section V. Although this may be done simply in a non-cooperative network scenario, the sophisticated cooperative adjustment of sensing times may also be provided.

A further increase of energy efficiency may be achieved by the optimisation of the decision threshold, λ , shown in Fig. 5 which is used for the differentiation of decisions reported in soft- and hard-decision scheme. However, taking into account network cooperativeness, joint optimisation of the decision threshold may bring even higher a gain than the one obtained with individual (distributed) optimisation.

2) *Branch B*: Another possibility of introducing energy efficiency in CSS comes from the observation that there may be some nodes in the network which bring a marginal profit to the overall detection performance. Therefore, it is beneficial to lower the number of active nodes when they encounter bad channel conditions in the sensed channel or their sensing results are highly correlated with other (neighbouring) sensors. In such a case, these nodes offer a relatively small added value to the network EE metrics at the high cost of consumed energy. Thus, it is good to reduce the activity of some nodes. One may pose two following questions: *i*) selection of which nodes is the most beneficial, and *ii*) how to organise such a selection?

Dozens of authors have tried to ask the first question and find the optimum sets of selected nodes for various topologies and network configurations. The general rule is that it is optimum to select the sensing nodes on the basis of the signal-to-noise ratio (in the primary- to secondary-user link) [69]. However, the possibility of proper selection depends on the radio propagation conditions. It is not always possible to estimate an SNR accurately. Moreover, the transmission of all approximation results consumes much of the spectral and energy resources.

As the nodes are recommended for active participation in the sensing process, the following cases can be considered. First,

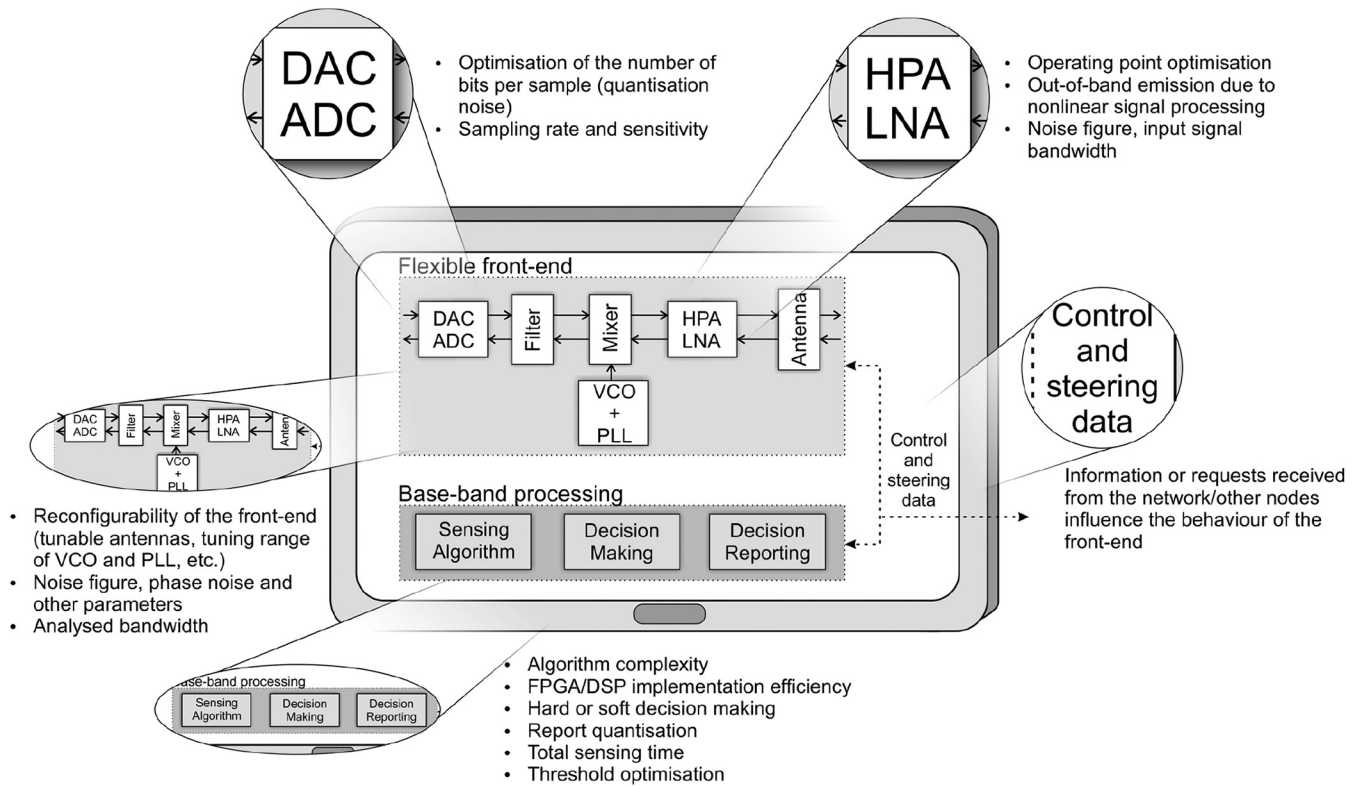


Fig. 11. Energy saving areas in single node.

a subset of nodes from a group can be *selected*, i.e., be active in the sensing phase, while the rest turns to the *sleep* mode, and does not sense the frequency-band occupation, nor report any sensing information. The nodes may also be *censored*, i.e., may sense the licensed signal but have no reporting rights, i.e., do not report local sensing observations to the other nodes. Although a limited number of active nodes lowers the overall energy consumption, the node selection process may be inefficient energy-wise, or incorrect (if wrong nodes are selected for a sensing or sensing-and-reporting group).

3) *Branch C*: Further energy savings are possible if an efficient fusion of sensing messages is considered. For example, for the hard decision reporting mode, it is possible to adapt the decision fusion threshold on the basis of the observed wireless channel conditions (e.g., [119]). When soft or quantised-soft metrics are taken into account, further energy efficiency improvements are possible.

4) *Branch D*: The consumed energy may also be limited by efficient network organisation. Clearly, if the amount of interchanged steering messages in CSS is high, the energy devoted to this process should definitely be considered more carefully. Thus, the question about cooperation gain and associated overhead in the centralised and decentralised CSS topologies is crucial. Recently, much attention has been given to the topic of relaying technologies. One may find papers tackling the problem of how relaying is energy efficient, and when it is more beneficial (in terms of the saved energy) compared to direct transmission (e.g., [120]). Apart from relays, the other solution for energy efficiency in Branch D is a cluster-based one.

Please note this is one of possible classifications. An other would be a division according to the provided sensing

procedure parts. In Fig. 10, one can see an illustrative diagram of cooperative spectrum sensing. Here, energy efficiency may be introduced for each state, e.g., in “Local Decision Reporting”, energy may be saved by a proper selection of cooperating nodes or a neat network organisation. Note that many solutions described in that work introduce energy efficiency by combining two or three different *branches* shown in Fig. 10.

V. ENERGY SAVINGS IN LOCAL SPECTRUM SENSING

Energy-efficient design and algorithms are possible inside a sensing device. The aforementioned energy-efficient hardware (e.g., power amplifiers) may significantly reduce the power consumption. This may also be done if the spectrum sensing procedure is optimised. The first possibility is to reduce the number of acquired samples or, in other words, reduce the sensing time. Although the number of samples may theoretically always be reduced when the desired sensing reliability should be achieved, the minimum number of collected samples can be found for each sensing method. Regardless of the number of samples required for further processing, specific features of the applied analogue elements in the transceiver front-end have to be defined (Fig. 11). For example, one of the key contributors to the energy consumed by the terminal front-end is the power amplifier. Optimisation of its operating point also optimises its energy efficiency, and can lead to the reduction of nonlinear effects (like intermodulation products) which are not allowed in practical systems, especially in the context of cognitive radio. Moreover, beside the total number of samples needed for the sensing algorithm, the frequency of their collecting,

as well as the number of bits used to represent each sample influence the energy consumed by the analogue-to-digital and digital-to-analogue converters (ADC/DAC). For example, in [121], it has been shown that for a given architecture, the power consumed by a DAC varies from 10 nW achieved for the sampling-frequency of 1 kHz to 26 μ W at the 10 MHz sampling frequency.

Let us also observe that the sensing nodes should be able to operate in various frequency bands. Thus, all the necessary elements should be reconfigurable, and very often such tuning possibility is provided to the user at the cost of complexity, thus, higher energy consumption. It is then even more meaningful to optimise the energy consumed by the wireless terminal front-end. The analysis of the energy consumed by particular elements of the radio transceiver front-end can be found in [122], [123]. In particular, in [123], the total power consumption at 1.1 V for various system architectures (GSM, DVB-H, WCDMA, MIMO WiMAX and MIMO WiFi) varies from 60 W to 230 W.

Note that the number of required samples strongly depends on the selected sensing algorithm. For example, cyclostationary-based detection involves hundreds or even thousands of samples in order to observe periodicity in the sensed signal. In energy detection, this number may be reduced even to one sample, however, too small a number of samples (or too short sensing time) leads to performance degradation. Therefore, there is a trade-off: on one hand, lowering the number of samples leads to detection degradation, while on the other, it guarantees lower power consumption, and results in a higher throughput because more time may be devoted to SU's data transmission (for more details see Sec. III-A).

Moreover, the selection of the spectrum sensing algorithm can also impact the energy consumption by the radio-frequency front-end. The application of the pure energy-detection algorithm entails a high dependency of the final sensing decision on the noise variance. Thus, in order to improve the reliability of detection, the sensitivity of the device should be as low as possible, also meaning that the resultant noise figure or the impact of the phase noise should be minimised. It also means that the bitwise representation of each sample should be possibly high in order to minimise the impact of the quantisation noise on the sensing procedure. Furthermore, the application of more complex algorithms typically results in more sophisticated hardware realisation (using FPGAs or a digital signal processor), and this entails a higher power consumption by these chips. An interesting discussion on the FPGA implementation of selected spectrum sensing algorithms can be found in, e.g., [124], [125].

Discussion on sensing time optimisation may be found in [126], where it is shown that there exists an optimum sensing time which guarantees a maximum throughput. Interestingly, the sensing time is similar irrespective of the transmitted power level. In [127], the sensing time was analysed from the perspective of time per bit. It was proven that the longer the transmission time related to one bit, the higher the energy efficiency. However, when the circuit energy in a realistic implementation is taken into account, there exists an optimum point of time per bit for which the energy per bit is the smallest.

However, in such an optimum point, the delay constraint has not been taken into account [128]. Finally, in [129], a sensing time allocation scheme for two PUs is proposed, while in [130], a neural-network-based optimisation is delivered.

It has to be underlined that the individual optimisation of the sensing time may affect the transmission throughput performance. In Sec. III-A, we described the idea of the joint optimisation of the sensing time and spectrum access.

Another possibility of increasing the energy efficiency is to optimise the sensing threshold. In [131], the threshold for energy detection is optimised. The algorithm in its two versions requires instantaneous or averaged SNR and it has been proved that the total energy may be reduced, and is supported by sensor-selection in the first step.

Another threshold optimisation scheme is described in [132]. Here, the optimum threshold for energy detection has been found for a cooperative network in which the nodes send their binary decisions to the fusion centre adopting the OR-rule. This solution also proves that a combined use of energy efficient *branches* improves the overall efficiency. Thus, the energy in CSS may be reduced for a single device, but a higher number of possibilities is opened when the energy is saved in a cooperative way.

Finally, let us note that in the context of the relaying nodes (or even cluster heads), the above discussion has to be slightly extended to the detection process performed before the signal is forwarded. Thus, the energy devoted to signal detection (and potentially decoding followed by re-encoding) has to be considered.

VI. NUMBER OF COOPERATIVE NODES

The strength of the cooperative solutions lies in the diversity of node locations and thus, in the variety of channel conditions and spectrum observations. Thus, the effect of faded and shadowed signal in one place for a single node is mitigated if a number of nodes cooperate in order to make a global decision. However, too high a number of the sensing devices may not bring relevant detection improvement while still consuming a high amount of power. Therefore, many authors try to find the optimum number of cooperating nodes and/or give a recipe for proper selection. For example, in [133], the number of nodes is optimised under specific time constraints. It is assumed that the duration of a frame and its transmission part is fixed. The general conclusion is that there exists an optimum number of nodes that maximises the energy-efficiency. Moreover, the longer the time for sensing and reporting, the higher the number of nodes maximising the energy-efficiency.

The optimum number of nodes may be selected in many ways. One may classify three main approaches. In the first approach, known as *node selection*, a subset of SU nodes is not allowed to sense the spectrum and report relevant sensing information (local decisions on spectrum occupancy). These nodes turn to the sleep mode and thus save energy. The licensed signal is sensed only by another disjoint subset of nodes. Unlike in node selection, in a technique called *censoring*, all nodes are allowed to make sensing observations but some of them cannot report this information. In this technique, it is possible to censor

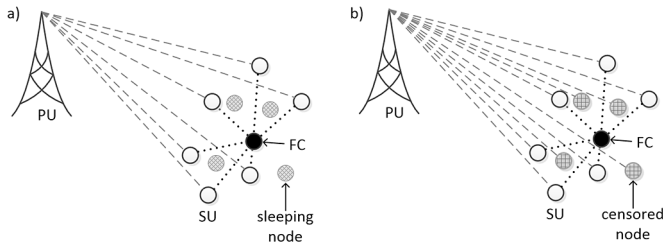


Fig. 12. Model of centralised CSS system where node selection (a) or censoring (b) is applied.

nodes during the sensing process, i.e., on the basis of up-to-date local sensing results. These two situations are illustrated in Fig. 12.

Naturally, lowering the number of active nodes may degrade the overall probability of detection. Thus, although a lower number of nodes decreases the number of transmitted messages with reporting information, the power devoted to one link is higher due to the longer distance to its neighbour when the decentralised scheme is used. Moreover, the energy savings nearly always have an associated cost. One is the degraded performance through the decreased global probability of detection or increased global probability of a false alarm. The other is that the sensing and reporting nodes have to be somehow selected and informed about their selection. The additional messages and processing cost should also be taken into account, as they may reduce the energy efficiency of the algorithm.

A. Node Selection Algorithms

The basic assumption in lowering the number of active nodes is to maintain the overall performance. This definitely depends on the criterion used for the selection of nodes in order to reduce the energy with an acceptable and possibly minimised cost of performance degradation.

The first criterion of efficient node selection is the observed signal-to-noise ratio. In [69], the authors propose such a selection and claim the solution is the optimum one. In this algorithm, the optimum value of Q_f (or Q_d) is found by the selection of an optimum number of nodes with the highest signal-to-noise ratios. However, for the proposed algorithm, up-to-date information about the nodes' instantaneous SNR is needed, and has to be delivered to the fusion centre, while the fusion centre has to receive the information from every SU in the network. Variable channel conditions induce SNR variations that must be dealt with, for example, by periodic updates of the estimates of the SNR for each node. It was proven in the above-mentioned paper that the appropriate selection of only 19 out of 200 nodes provides promising results: under the CFAR with AND rule, the global probability of detection is improved from 92.04% to 99.88% and much energy for the decision interchange is saved.

Another algorithm based on the SNR criterion has been described in [134]. In this work, the secondary user with the highest SNR is chosen in the first iteration. Next, every other node compares the quality of its link to the fusion centre with the quality of its link to the formerly selected node and from the

formerly selected node to the fusion centre. If a node determines that its own link is less reliable, then it joins the group of nodes experiencing the highest SNR. Otherwise, the next highest-SNR node among ungrouped nodes is selected, and then the procedure of comparing link qualities and grouping is repeated until all nodes are grouped. Again, reliable information about the SNR ratio is demanded. Moreover, the energy efficiency of the scheme depends on the number of selected group-heads: the lower the number of selected heads, the higher the energy efficiency, up to 76%.

Another interesting SNR-based selection algorithm has been proposed in [135]. In this work, the nodes are classified either as leaders or followers based on the received SNR. The leading nodes have good detection performance and are allowed to sense the PU signal and broadcast their sensing information. The following nodes are considered unreliable due to a low SNR, so they do not broadcast their decisions, but rather wait for the broadcast packets from leaders. Thus, only the reliable information is broadcast. In addition, the information sent by the leaders is rather limited, only consisting of the PU-presence information. As a result, the approach proposed in [135] leads to low overhead information. However, the identification of nodes with the highest SNR is challenging, as it must rely on the presence of the PU during the training (measurement) periods.

Typical SNR-based node selection similar to the aforementioned work [69] was described in [136]. In this paper, an adaptive double-threshold method was also introduced, and connected with noise uncertainty. The presented results showed, as emphasised in [69], that it is beneficial to select only a minor percentage of nodes with the highest SNR, thus guaranteeing an optimum global detection probability and important overhead reduction.

Although it was underlined that SNR-based node selection brings significant gains, in the literature, one may find another promising criterion of node selection. It is the correlation-based node selection algorithm which is based on nodes' decisions about PU presence. The idea is to select nodes whose sensing observations are uncorrelated. It relies on the assumption that the selection the uncorrelated nodes should result in a high detection quality, while significantly minimising the energy overhead spent in the network for reporting.

The idea of correlation-based node selection comes from [137]. The selection of nodes is proposed to be conducted with the use of a correlation measure computed by the nodes. Starting with a randomly selected node, the nodes, one after another, compute the correlation of their own sensing decisions with the decisions of other nodes pair by pair. If the sum of correlation coefficients is high, the node becomes inactive and is removed from further correlation computing. The correlation measure in [137] involves location of nodes and distances.

Unlike in the aforementioned article, in [74], the correlation is based only on sensing messages received from nodes in the network. A randomly selected node broadcasts its sensing observation, while every other node listens to messages received from other nodes and calculates the correlation to that. If the correlation is above an assumed threshold, the node becomes inactive. Nodes which have a correlation below the threshold select a random delay and the one which has

the shortest delay reports its data. The procedure is repeated until there are remaining uncorrelated nodes. After the procedure is completed, all nodes may specify the global decision because during the procedure they got the messages from all uncorrelated (i.e., selected) nodes.

In [138], the correlation between nodes is delivered only on the basis of decisions made by nodes. In the algorithm, a number of sensing decisions has to be delivered to the fusion centre. The FC then derives a correlation matrix, and creates correlated groups of nodes according to the minimum correlation threshold. The information about the formed groups is sent to the network, and then, in each group of correlated nodes, a leading node is selected. In [138], the node with the highest detection probability is selected as a group leader.

In [139], basing on [138], a different selection of group leader has been proposed. It has been shown that in a mobile scenario, the formed group of nodes may quickly become obsolete. Thus, the leader selection metric considering both node mobility and sensing performance has been proposed.

B. Censoring

In the censoring algorithm, all nodes sense the spectrum band, however, only some of them are allowed to report their observations. Thus, the energy is saved during the reporting stage when some nodes do not transmit their observations. Moreover, censoring is more reliable than node selection because the censored nodes are selected after each sensing period.

An example of an algorithm based on censoring is introduced in [137]. In a network consisting of N nodes, all nodes are grouped to the active set at the beginning of the algorithm. After the selection, only X nodes may remain in the active set, while the rest is moved to the passive set, and that includes all nodes that are not allowed to vote for the global decision. In order to make a proper selection, the correlation measure is computed for pairs of nodes in the network. Then, the node with the highest summed correlation with the remaining sensors is removed from the active set and moved to the passive set. The *correlation measure* used in [137] is based on the positions of nodes and associated positioning uncertainty.

In [140], the censoring scheme was optimised. The authors put censoring into the global cost-function of sensing, and proposed a solution for the selection process based on the double-threshold energy detection. The sensing decision is censored if the collected energy of the PU signal falls in between two thresholds. Under some constraints (the assumed global probabilities of false alarm and detection, and the OR fusion rule) it was shown that the censoring probability for an average node saturates rapidly and is not dependent on the number of cognitive radio nodes, leading to a lower consumed energy. The scheme can be easily adopted in a practical network.

The idea of censoring was further investigated in another article by the same authors [141]. There, a similar model of cooperative network was proposed, where censoring and sleeping are adopted under the constraints of minimum detection probability and maximum false alarm probability. Moreover, the authors provided a detailed analysis of energies consumed

in the network. They compared two cases: A) when the sensing and transmission energy are equal, and B) when the sensing energy is a minor part of the transmission energy. It was shown that the censoring rate in case A) is lower than in case B), and that the rate of turning to the sleep mode is higher in A) than in B). Moreover, the proposed solution was adopted in the real ZigBee transmission network. It was also proved that it is possible to find the optimum censoring and sleeping rates and using these values may result in significant energy savings.

Finally, in [70], the idea of sequential censoring was introduced as a combination of traditional censoring with the sequential algorithm. Again, the authors found the optimum values for censoring thresholds both for the OR and AND rules. It was shown that for low-energy radios, sequential censoring outperforms regular censoring in terms of energy efficiency. Moreover, the AND fusion rule guarantees a lower energy consumption for medium values of PU presence probability (not higher than 0.8) than for higher values.

C. Voting Schemes

Note that it is not always possible to properly estimate the signal-to-noise ratio. Moreover, it may not be energy-efficient to interchange messages in order to find the specific SNR value. Therefore, voting schemes have been proposed, based on the observation that the conclusion of own and global decisions may be based on an SNR-like metric.

The first representative of voting schemes is the so-called *Confidence Voting* [142], in which the nodes build reliability-related measures. The idea is to limit unreliable decision transmissions. Every node is obliged to compute a confidence metric. In the hard decision scenario, the local and global decisions are collated. In the case of coincidence, the confidence metric is incremented, otherwise it is decremented. After the training period, in which the metrics are computed, only the nodes with the highest confidence metrics are allowed to report their decisions to the fusion centre. The authors claim that up to 40% of energy may be saved when using their algorithm.

The Collision Detection scheme, presented in [143], is based on node selection with the highest correctness measure. The measure notifies the number of a node's correct decisions when the global false decision is that the PU is not present. The nodes with the highest correctness are selected and involved in cooperative sensing.

The schemes based on voting have the advantage of being applicable in scenarios when there are no periods in which the presence of the PU is known in advance. However, these schemes have a major drawback. As they rely on the opinion of the majority, if most of the secondary users face bad channel conditions in their links between the PU and themselves, more confidence goes to unreliable nodes. As a result, the decision obtained in confidence voting may be worse than in the traditional scheme. Moreover, the voting schemes are not robust enough in the case of the presence of malicious SUs. A malicious SU is the one that sends untrusted decisions, that makes the decision taken in the cooperative network unreliable [144].

In [145], the authors propose a scheme that combines energy efficiency and sensing performance in node selection. The

scheme introduces a cost function that favours nodes with the lowest sensing and decision-transmission energy usage among those satisfying the quality of detection constraint. Furthermore, energy efficiency is increased by introducing the *Decision Nodes*, each acting as the collector of sensing results from a set of selected nodes, determining a common decision and sending it to the fusion centre. The scheme requires a knowledge about the nodes' signal-to-noise ratios and the distances between each node and the fusion centre in order to operate, leading to a significant control overhead.

In [146], the *multi-channel aware* algorithm is proposed. The nodes with the lowest energy consumed by sensing and reporting, and the lowest number of channel switches are selected. Moreover, the nodes experiencing higher SNRs (in the links between the PU and themselves) are chosen due to shorter demanded sensing time. There, the nodes' selection is performed with an occurring delay constraint. The algorithm results in low energy consumption.

VII. FUSION RULE

As mentioned above, the decrease of energy consumption is possible if the fusion scheme is appropriately optimised. In Sec. II-B, it is stated that there exist two basic types of decisions transferred from nodes to FC: hard- and soft-decisions. We describe the main objectives and propose some references: [56], [59]–[63], [71]. This essentially determines the effectiveness of decision fusion. The works presented below deal with the optimisation of the fusion rules, which for the hard-decision reporting are as follows: OR, AND and the majority rule (known as the k -out-of- N rule).

In [64], the authors analyse the energy consumption and detection probability of three fusion rules under three parameters: frame length for each rule, the number of nodes and SNR. The results show that for the critical set of conditions (very short frame length, substantial number of nodes, low SNR), the Equal Gain Combining rule outperforms the Likelihood Ratio and Maximum Ratio Combining rules [64].

A vast majority of authors make the assumption that the reporting channel is ideal and error-free. This is, however, impractical in an actual CSS network. In [62], one may find an interesting analysis of the fusion rule performance where the reporting channel is not ideal. Hard decisions and soft decisions have been taken into account. It has been proven that the soft decision combination is more robust to channel impairments. However, the work has not covered the topics of the associated complexity and transmission overhead.

In [119], the authors propose optimisation of number k in the k -out-of- N fusion rule together with energy detection threshold optimisation with the aim of energy efficiency maximisation in a network of cooperating nodes. The presented results show that the joint optimisation of k and the decision threshold may lead to energy efficiency up to 2 bits/Hz/Joule for different SNRs.

The *Adaptive Counting Rule*, which is in fact an optimisation of the majority rule, is proposed in [147]. There, a cooperative network of n SUs is considered. The adaptive rule is adopted in the hard-decision fusion scheme. It optimises the number of sensing SUs k declaring the presence of the primary signal. It is shown that the optimum minimum value of k depends

on the value of the calculated correlation of the nodes' decisions, as well as the number of detectors in the network and their detection performance. The authors also propose a continuous mechanism of selecting optimum k value. However, the work lacks information about introduced overhead needed for the calculation of the optimum k .

An energy-efficient algorithm connecting the majority fusion rule with the sequential algorithm can be found in [148]. A two-stage algorithm was proposed, where in the first (coarse) stage, sequential sensing is applied with the modified majority rule. The stage is finished if more than a half of sensors declare the same decision. If the condition is not fulfilled, a fine stage is applied with the traditional energy detection algorithm and the accustomed majority rule. The proposed scheme may help in saving energy by up to 30% for low SNRs and 60% for high SNRs for most common settings.

In [149], optimum linear cooperative sensing is presented. The authors propose a method based on the combination of test statistics from the local nodes, instead of the full energy-detection reports with associated transmission overhead. The authors have conducted different optimisations for cases with different values of detection and false alarm probability. They claim that the optimisation of the fusion rule has to be provided for specific cases, e.g., for cases with a low possible false alarm rate and with a low detection rate.

VIII. EE NETWORK ORGANIZATION

In this section, we present energy-efficient network-organisation methods that can be applied in CSS networks, allowing for energy-efficiency improvement. We do not address the topic of Medium Access Control with its details about spectrum access contention. Some details about spectrum access optimisation can be found in Sec. III-A and in the comprehensive surveys: [89], [90].

In [120], the authors analyse the benefits of relaying the sensing information. They propose a two-stage algorithm consisting of the broadcasting and the relaying phase. The authors adopted the Bellman-Ford algorithm which for a given network-graph minimises the cost of transmission from a source to a sink in a distributed manner, taking the required transmission powers into account. The *optimum cooperative route* is found in an iterative way by exchanging a number of messages between the nodes, thus making the algorithm relevantly complex.

The same Bellman-Ford algorithm was used in an earlier work [150]. Here, two solutions are proposed: first, minimising the power consumption in the route, and second, minimising the power under the constraint of achieving the assumed throughput. However, in this work, the power consumption is optimised only for single ways, not for the whole network. Moreover, the scheme adopts a single-relay cooperation model due to an increased complexity for a higher number of relays, although in that way simplifying the solution. The presented performance analysed for linear and grid networks shows significant power savings compared to the shortest-path algorithm.

An energy-efficient network with relays is also proposed in [76]. A wireless network of transmission pairs with the use of relays is analysed. There, it has been stated that for each transmission pair, one relay had been considered. With the

introduction of virtual relay, the transmission mode selection has been simplified (direct vs cooperative). Then, the proposed iterative solution optimises the power allocation levels in order to find the minimum of the total power consumed in the network. However, the authors introduce a doubtful performance metric which is the *transmission reliability*, and adopt a limit of one relay per one transmission link. Finally, they present promising results of obtaining fairness in the network and solve the max-min fairness resource allocation problem. This fairness depends on the assumed transmission powers and the so-called transmission reliability being the probability that SNRs between source, destination and relay nodes follow a given criteria.

A similar single-relay scheme was proposed in [77]. The proposed protocol is based on request-to-send/clear-to-send (RTS/CTS) messages sent by a source and a sink, and on the contention phase in which relaying candidates compete. The contention is organised in a manner minimising the signalling overhead. Then, two solutions are proposed: minimisation of the total transmission energy and maximisation of the network lifetime. The presented results prove that direct transmission is outperformed by the proposed minimum-energy scheme, in terms of the consumed energy per packet and the network lifetime.

A similar cross-layer distributed algorithm can be found in [78]. As in some previously described schemes, here, the best relaying node and allocated power optimisation is found. Moreover, the authors present not only the prevailing min-energy total consumption but also introduce an interesting transmission rate-power trade-off. It was shown that under the energy minimisation constraint, the cooperative scheme outperforms the non-cooperative one, however, in the latter case, the gap in the defined utility-energy trade-off between cooperative and non-cooperative solutions is smaller.

In [151], the authors propose an energy-efficient algorithm with the assumption of partial channel information. The authors prove that their heuristic approach results in significant energy savings, and additionally, it is computationally efficient.

In [152], an optimum routing strategy in a multi-hop network is proposed with identifying important delay constraints. The authors adopt a scheme where mutual information is accumulated after each packet transmission. Every node may send its information at any time, after it receives the full packet composed of messages sent by other nodes. The authors propose a routing scheme minimising the total energy consumed in the transmission link under the constraint of a given delay. They show that this problem may be solved with the use of a greedy algorithm similar to the minimum delay routing. To this end, they introduce two heuristic algorithms presenting promising performance and underlining the fact of a limited overhead.

The authors of [153] presented the analysis of energy efficiency in a system with one-way or two-way relaying. Although the proposed three-node scenario is quite simple, with a single relay positioned exactly between the transmitting and the receiving node and with a simplified channel model, the presented results are interesting. Generally, two-way relaying is more energy efficient than one-way relaying, since symmetrical transmission is considered. However, it is shown that relaying

is not always more energy-efficient than direct transmission. If the channel attenuation is moderate, direct transmission is recommended. Otherwise, relaying is beneficial. Moreover, some results related to the considered circuit power show that for a non-zero transmit, receive and idle power, energy efficiency has its non-zero maximum for medium spectral efficiency.

In [154], the authors presented an optimisation of energy consumption in a network with some selfish nodes which may not be willing to relay other transmissions. The authors unveiled the scheme of forming partnerships with selfish nodes under the condition that no central entity is employed. Finally, the presented results proved that the proposed bargaining technique guarantees about 50% of energy efficiency when compared to a centralised random scheme without selfish nodes.

In [155], a cooperative beamforming scheme was described. The authors analysed the energy efficiency of direct and cooperative schemes in the function of distance between the communicating entities. It was shown that the adaptive direct scheme, in which the transmission power is neatly adjusted, outperforms the cooperative schemes for transmissions on short distances (up to ca. 150 m to 200 m). For larger distances, cooperative relays are more energy efficient, however, the larger the distance, the higher the optimum number of the relaying nodes.

In [156], a scenario without any relays was analysed. The authors took two known algorithms into account: the *gossiping* and the *random walk*, and proposed improvements in order to decrease the information overhead in the network (resulting in an energy-efficiency increase). In the gossiping scheme, every node transmits information to a randomly selected node. In its enhanced version, it is done only if the usage pattern of a given band changes. In the random walk algorithm, only some nodes communicate to randomly selected neighbours while in the incremental version of it, the procedure is started if an information update occurs. The authors present the results which prove that the proposed enhancements decrease the number of overhead information up to 2.5 times. However, the definition of update is dubious, as well as a number of collisions which may occur in the presented schemes, e.g., when a node cannot transmit in the same time as it receives the information signal.

In [157], the trade-off between sensing performance and its energy efficiency was discussed. The authors present a scheme in which a larger number of samples increases the detection probability. However, if the minimum detection probability is satisfied, the energy consumption increases linearly with the number of samples. The authors analysed the scheme with a simple amplify-and-forward scheme. The presented analysis shows that it is possible to find the optimum pair of the amplifying gain and the number of samples in order to get the best trade-off between sensing performance and energy efficiency.

Another solution which is not based on the idea of relays is presented in [158]. Here, two heuristic models were presented in order to optimise network utility and energy efficiency. In the network model, the authors assume several licensed signal spectra present in several fragmented frequency bands. They adopt the economic concepts of social welfare and net revenue to the communication network scenario. There, it is proved that the optimum trade-off between energy consumption and

social welfare may be found by allowing for the interference-dependent competition for links and source nodes.

The next solution proposed for a network with multiple-license signals was described in [159]. Here, the proposed solutions aim at maximising the energy efficiency and transmission rates. It is achieved by allowing the node to use vacant channels with the upper-bounded transmit power, and to transmit multiple packets in one transmission.

In [160], the *cluster-and-forward* scheme is presented, where the nodes are dynamically put into cluster groups. In each group, a node with the best channel gain is selected as the Cluster Head. Then, it collects local decisions from cluster members, forwards them to the Fusion Centre and, in order to improve the energy efficiency, also serves as the fusion centre for cluster members. In the paper, it is shown that for a given number of nodes, there exists an optimum number of clusters for which the amount of saved energy is the most significant.

A similar clustering scheme is described in the aforementioned article [142]. There, the total transmit energy in clustering and broadcasting is compared, proving that the clustering may provide significant energy savings due to shorter transmission links. Moreover, there exists an optimum number of clusters for a given number of nodes which guarantees most of the energy savings. A similar conclusion has been drawn in [161]. The transmit energy may be reduced compared to the traditional scheme, especially when the transmission takes place over large distances (1000 m and more).

A cluster-based efficient MAC protocol was described in [162]. First, a MAC mechanism was proposed in order to reduce the number of collisions. Then, a channel sensing scheme was designed to reduce the total consumed energy. It was proved that there is a relationship between the energy efficiency and the number of sensed channels, and the best overall performance is guaranteed by cluster-based sensing for three channels.

Walid Saad *et al.* propose in [163] an interesting attitude to cluster formation. Although usually clusters are formed during a centralised procedure where the key role is performed by the central entity, in the cited work, node collaboration is provided in a distributed way. Two coalition-formation approaches have been proposed. In the first one, under the false alarm requirement, the nodes may form clusters (coalitions) by pairwise negotiations between them, followed by the sequential *merge-and-split* procedure in order to maximise detection probability and keep the false alarm rate sufficiently low. The second one aims at forming coalitions under the minimum-detection constraint, while keeping the false alarm rate at a given level, thus guaranteeing the achievement of detection probability with a minimum overhead. This is provided by forming *minimal winning coalitions*. One may find promising results where the miss-detection rate has been significantly reduced and the detection rate relevantly increased for the first and second algorithms, respectively.

IX. FINAL CLASSIFICATION, CONCLUSIONS AND CHALLENGES

In the previous sections, the ways to increase the energy efficiency in CSS have been analysed. Here, we summarise

the discussed methods in the form of a table, which should guide the reader through the potential papers for further reading. The analysis of Table V allows us to identify the best ways for global energy consumption minimisation in CSS depending on various optimisation constraints. Following the overall paper structure, we have split the table into four separate (yet mutually related) parts which reflect the four branches defined in Sec. IV. Within each part, we have identified the key CSS aspects which can be subject to optimisation. However, the key challenges in accurate energy consumption modelling and optimisation for CSS networks arise from the fact that there exist a great variety of elements that have to be considered in the optimisation process. Optimisation of one parameter (e.g., selection of cyclostationarity-based spectrum sensing instead of energy detection in each node) increases the reliability of the decision made in one sensor at the expense of computational complexity. From the perspective of the whole network such a modification can result in relatively significant changes in energy consumption. Moreover, various factors that influence the energy consumption are mutually dependent, which makes the analysis even harder.

A. Dependency Matrix

In order to identify the most promising optimisation areas, a matrix has been created that shows the key relations and dependencies between the particular CSS solutions (see Tab. II). Based on the classification in Fig. 10 and its detailed description in Tab. V, we compare fourteen CSS features which are denoted according to the branch each feature belongs to. Each cell in the dependency matrix defines the mutual influence of two selected features, for example, the cell in column A3 and row B2 represents the relation between message quantisation and node censoring. We have assumed that each two features can be highly correlated (dependent) when the change of one of them significantly influences the second one. Thus, this *dependency parameter* can vary in the range from 0 to 1, where 0 means no correlation. Each cell uses colour coding, meaning that the darker the colour, the higher the correlation between the parameters of a row and a column.

Before we start an analysis of this dependency matrix, some additional comments have to be made. First, we have identified fourteen specific CSS features, however, that selection is a matter of classification. One can easily define other sets of features allowing for drawing conclusions. Second, in the discussed example, we have assigned the dependencies between the features somehow arbitrarily, based on our observations and overall assumptions. One can identify more entries to the matrix records once the assumptions are modified. However, in order to achieve precise results, one needs to define at least a generic metric or – at best – a mathematical relation between each pair of features. This is the key challenge of the proposed approach, as such a metric could be hard or even impossible to define.

Having in mind all of these limitations (i.e. the matrix is case-dependent, it relies on user experience and knowledge etc.), we do think that a coarse analysis of such a matrix can shed new light on the overall understanding of the CSS process and the existing relations between its particular phases.

TABLE II
DEPENDENCIES BETWEEN THE KEY CSS PARAMETERS

Parameter	Description	A1	A2	A3	A4	A5	B1	B2	B3	C1	C2	C3	D1	D2	D3
A1	Reduction of the number of collected samples														
A2	Application of an advanced, energy efficient front-end														
A3	Message quantisation														
A4	Optimisation of the decision threshold														
A5	Selection of a simpler sensing algorithm														
B1	Node selection														
B2	Node censoring														
B3	Voting schemes														
C1	Selection of the key fusion rule														
C2	Soft or hard reporting														
C3	Decision reporting														
D1	Energy-effective routing														
D2	Cross-layer solutions														
D3	Application of relaying nodes														

First, let us notice that by finding similarities between the rows (or columns), the most dependent features can be identified, meaning that in consequence, a dedicated optimisation function can be defined that considers these particular features. Second, as the darker entries in the matrix represent high dependency between the features, the brighter ones allow for the identification of such agents of CSS which could be optimised independently. In general, such an analysis can lead to the definition of sets of highly mutually-related features (note that these sets can overlap).

In order to deal with the problem of precise definition of dependencies between any pair of features, we proposed to apply the approach known from fuzzy logic, where the exact values can be intentionally replaced with some generic, descriptive definitions. In our approach, we propose five levels defining the degree of mutual dependency:

- Level 0 – no or very low dependency;
- Level 1 – low dependency;
- Level 2 – moderate dependency;
- Level 4 – high dependency, and
- Level 5 – very high or full dependency.

The number of levels, as well as their (fuzzy) meanings define the accuracy of the conclusions that will be drawn from it. Tab. II has been created using the five levels mentioned above.

In Tab. II one can notice that the highest correlation exists between solutions within branches B and C, it is substantial in branch D and moderate in branch A. Moreover, the solutions in the dependency matrix may generally be split into

two groups: in the first group, consisting of solutions from branches B, C and D, one may observe a rather high dependency between them (these are essentially cooperation-based solutions), while in the second group, consisting of solutions from branch A (these are single-node solutions), there is a much lower dependency with the exception of A3 highly correlated with branch C ideas. Therefore, the general observation is that the introduction of one energy-efficient cooperation-based solution (e.g., the number of nodes or acceleration of the fusion rule) highly affects the others. However, the optimisation of a single node's operations does not impact so much the energy-efficiency solutions of the other branches.

One can also notice that in our example, we have assumed absolute values of dependencies. However, much effort should be put into the precise definition of mutual relation between features, i.e., the modification of a certain feature can either improve or degrade the energy efficiency of an other feature. These aspects (i.e. positive or negative influence) could also be taken into account, providing new insights into the problem.

B. Rose Charts

As the dependency matrix (or, in some sense, correlation matrix) provides us with some statistical insight into the energy efficiency of CSS, the key challenge would be to define a detailed energy consumption model. However, as the full model of energy consumption in a whole CSS network would be highly complicated, we propose to identify the key relations between the factors that influence the total energy consumption.

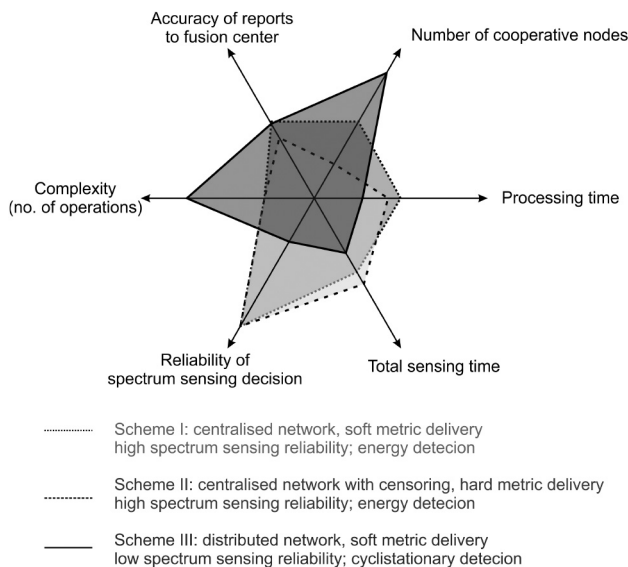


Fig. 13. Rose-chart for energy consumption comparison.

It can be achieved, to some degree, by the analysis of a rose chart with key factors assigned to separate axes, where each axis of this chart represents a different feature (criterion) considered in the analysis. All of the axes start at the same zero point, and it is important to precisely define the terms and units in which the selected feature (criterion) can be defined numerically. Once the numerical values are marked on the axes, a polygon can be created through the connection of all points, and the area of this polygon to some respect reflects the overall energy consumption of the considered system. An illustrative example showing the concept has been presented in Fig. 13.

Here, we have identified six criteria for evaluating CSS schemes:

- Accuracy of reports delivered to fusion centre;
- Number of cooperative nodes
- Processing time
- Total sensing time
- Number of operations executed in each node;
- Reliability of spectrum sensing decisions.

Similarly to the approach applied in the creation of filling in the dependency matrix, also in this case (the rose chart), particular attention has to be given to the accurate definition of the considered criteria (the rose chart axes). As we have already stated, there always exists some mutual dependency between any pair of the proposed criteria. Having this in mind, we propose to select them in such a way that the mutual dependency is minimised. In order to achieve this, one can utilise the dependency matrix presented at the beginning of this section.

Once the criteria (including their number) have been defined, the most important thing would be to precisely define the quantitative metric used for the numerical assessment of a criterion. For example, the number of cooperative nodes can be straightforwardly measured in terms of integer numbers, and complexity – understood as the number of operations executed in one node, can be measured in operations per second or flops¹.

Similarly, the processing time can be easily represented in the form of seconds. On the other hand, the numerical assessment of reliability of sensing decisions or accuracy of reports are not as straightforward. As one can consider various types of probabilities (e.g. probability of a false alarm, error probability, etc.), also other metrics could be proposed. Finally, it would be wise to translate such generic measures into concrete values of consumed energy. In other words, taking the total sensing time as an example, although it is measured in seconds, it should rather be analysed as a function of time that gives us even a rough approximation of the energy consumed.

As the definition of accurate functions describing the energy consumption in each criterion can be complicated, we again propose to apply the methodology known from fuzzy logic, as we have done previously. Thus, depending on the criterion, particular levels can be defined giving coarse estimations of their impact on the energy consumption. For example, complexity or accuracy of reports can be classified as very low, low, medium, high, and very high; analogously, the number of cooperative nodes can be low, medium and high. One needs to propose the mapping function that connects pure numerical values with their descriptive counterparts.

In Fig. 13, a comparison of three arbitrarily selected schemes is presented, allowing us to identify the key differences between them and to assess the key contributions to the total energy consumption in a CSS network in each case.

In schemes I and II, centralised networks of energy detection-based nodes are assumed. Thus, both schemes have a low complexity and a high reliability. However, in the second one, the nodes are censored and use hard-metric delivery, while in the first, soft metrics are used. Therefore, they have different accuracies of reports, as well as sensing and processing times. In scheme III, we propose a distributed network (with a high number of nodes) with cyclostationary detection. Thus, we have a short sensing time at the cost of high complexity and low reliability. The three presented schemes are simple proposals, however, one may observe the great variety of parameters that influence the total energy efficiency.

C. Exemplary Use Cases

In this section, we attempt to apply the proposed evaluation tools (i.e., dependency matrix and rose chart) to a very specific example. We have arbitrarily selected two different test scenarios (use cases), deriving from them the illustrative values of the parameters and choosing appropriate methods.

The first considered use case is a mobile dense network which is monitored by densely-deployed static sensors and, additionally, by mobile users who deliver the sensing results to a centralised fusion centre. The obtained spectrum occupancy information is used for updating global databases (REMs). Each static node periodically senses the spectrum based on a cyclostationary-features algorithm, while mobile users perform sequential energy detection. As the static nodes report the measured values to the fusion centre, the mobile users report their hard decisions, as well as their location. FC applies the majority rule to these reports, and: a) circulates the decisions to the affected users (e.g., a certain user may need to modify its

¹FLOPS stands for Floating Point Operations Per Second

TABLE III
DEPENDENCIES BETWEEN THE KEY CSS PARAMETERS IN MOBILE
DENSE NETWORK USE CASE

Par.	A1	A3	A4	B1	B2	B3	C1	C2	C3
A1									
A3									
A4									
B1									
B2									
B3									
C1									
C2									
C3									

transmission parameters), *b*) updates the REM, *c*) optimises the majority rule in order to guarantee a high detection rate and a low false alarm rate, *d*) switches certain sensing nodes on or off in order to minimise the utilised energy, while keeping the sensing performance unchanged.

In the second scenario, we consider machine-to-machine communication, where a set of dedicated sensors is deployed over a large area and used for monitoring the occupancy of data transmission in unlicensed bands (e.g., 5 GHz). The role of the CSS system is to detect the presence of any WiFi users, particularly considering the problem of a hidden node, and to provide some valuable updates to the operator who applied the Licensed Assisted Access (LAA) spectrum sharing scheme [164], [165]. The nodes create a mesh network with the ability to define clusters, cluster heads and routing rules. However, most decisions are made by the centralised fusion centre which receives information from the cluster heads only. FC can take the following actions: *a*) instruct the cluster heads to activate/deactivate sensing nodes, and to reconstruct the network, *b*) deliver information about presence to databases (or other entities from the 5G network architecture), *c*) instruct affected users (via cluster heads) about the measured activity of WiFi users. For example, when the total measured power in the observed unlicensed band is high, the 5G operator decides to switch off the LAA strategy and rely on the licensed band only. In such an approach, the decision about the measured interference should be reliable, but there are no strict requirements on the reporting time and periods.

In Tables III and IV, we present dependency matrices for the analysed use cases. One may observe that these dependency matrices are created on the basis of Table II. Each specific use case has given constraints, thus, a subset of EE methods cannot be used. Consequently, dependency matrices presented for use cases are created for a given subset of EE solutions. Now, one may define the possible directions of EE optimisation and observe the possible dependencies between features.

TABLE IV
DEPENDENCIES BETWEEN THE KEY CSS PARAMETERS IN
MACHINE-TO-MACHINE USE CASE

Par.	B1	B2	B3	C1	D1	D2	D3
B1							
B2							
B3							
C1							
D1							
D2							
D3							

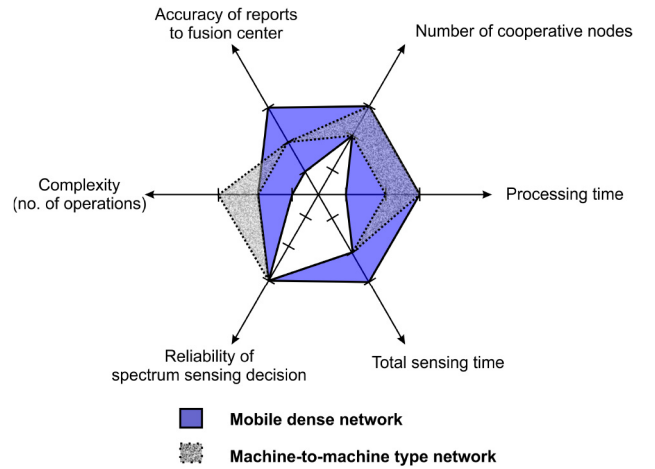


Fig. 14. Rose-chart for presented use cases.

The picture of EE optimisation may be amended by the observation of rose charts for the two considered use cases. In Fig. 14, one may note that the values on the axes may be strict (e.g., for reliability) or may be determined by the range. For instance, in a machine-to-machine communication use case where the local sensing phase could not be rearranged, the value for sensing time is strict (similarly for the FC reports on accuracy and reliability). However, the three remaining axes present features that may be optimised, thus, there exist ranges of values. The application of a specific routing scheme, which is a possible direction of optimisation, may affect the complexity and processing axes in the rose chart, and is correlated with the number of cooperative nodes. The selection of a given EE solution influences other solutions (the dependency matrix highlights this) and further affects other axes of energy efficiency (observed in the rose chart).

An other considered use case of a mobile dense-network presents a higher flexibility in EE optimisation than the machine-to-machine communication use case. Just one parameter is strict (reliability), while the five remaining ones may be subject to optimisation. This goes in line with Tab. III, where

TABLE V
CLASSIFICATION OF ENERGY SAVING APPROACHES

		Feature	Pros	Cons	Reference
Branch A	Energy reduction in the (local) spectrum sensing phase	A1. Reduction of the number of collected samples	• Less data to send and process • Shorter sensing time	• Potentially less reliable decision • Good candidate only for strong PU signals or lack of transmission	[126]–[130]
		A2. Application of an advanced, energy efficient front-end	• Better energy utilisation • Front-end adapted to current requirements	• Mass-production of electronic chips limits the degrees of freedom in energy-efficient front-end design for spectrum sensing purposes • More advanced chips typically leads to higher prices	[114]–[118]
		A3. Message quantisation	• Reduction of the data volume to send (reduced traffic in the CSS network) • Optimised number of quantisation levels can guarantee the agreed level of QoS in CSS	• Insufficient number of quantisation levels reduces the reliability of global decisions made in CSS	[57]–[63]
		A4. Optimisation of the decision threshold	• Reduction of the processing load • Optimisation of sensing time	• Wrong threshold selection influences CSS reliability	[131], [132]
		A5. Selection of a simpler sensing algorithm	• Reduction of processing load in each sensing node • Reduction of energy consumption in each node	• Poorer (e.g., less reliable) spectrum sensing algorithm can breach the assumed QoS	[124], [125]
Branch B	Optimisation of the number of cooperating nodes (see Tab. I)	B1. Node selection	• Reduction of the sensing node number • Sleeping nodes do not consume energy • Proper selection of the reporting node can improve the quality of the reported message	• Possible detection probability degradation • Proper selection of reporting nodes possible when additional data is acquired (e.g., nodes' SNRs)	[69], [133]–[139]
		B2. Node censoring	• Reduction of the reporting node number • Censored nodes do not consume energy for reporting • Faulty or even malicious nodes can be censored	• The censoring process may be faulty leading to wrong decisions • The need for duly node selection for censoring or turning off	[70], [137], [140], [141]
		B3. Voting schemes	• Easy usage of the procedure • Limited number of steering messages • Does not demand much additional data (nodes' SNRs, locations etc.)	• Faulty or even malicious nodes may affect performance	[142]–[146]
Branch C	Proper selection and application of fusion and decision rules	C1. Selection of the key fusion rule	• The number of reports delivered to the fusion centre depends on the selected fusion rule • Fusion rule can be selected depending on the required quality	• Every change of the fusion rule has to be reported to the cooperative nodes • The efficiency of the fusion rule has to be monitored	[56], [79]–[81], [119], [147]–[149]
		C2. Soft or hard reporting	• Depending on the situation, soft or hard reporting can be selected • Adaptation of the traffic related to data reporting	• Trade-off between the accuracy of each report and amount of data volume needed to deliver	[56]–[58]
		C3. Decision reporting	• Reduction of the traffic	• Reduced accuracy of information delivered to the fusion centre (compared to soft/hard reporting)	[64], [82]–[85]

TABLE V
(Continued.)

Branch D	Selection of energy-efficient network organisation solutions	D1. Energy-effective routing	Selection of the routing scheme that minimises the consumption of energy	Application of energy effective routing requires access to detailed information about the environment	[120], [150]–[163]
		D2. Cross-layer solutions	- Possible cooperation between solutions applied in separate OSI layers, e.g., routing algorithms can consider physical-layer constraints - Increased degree of freedom in system design	Increased adaptation and flexibility in system design typically results in the application of advanced (thus complicated and more energy-consuming) algorithms	[78], [158]
		D3. Application of relaying nodes	Reduction of the transmit power (thus overall interference level observed in the system) of the reporting node due to shorter distances	- Increased energy consumption in relaying node due to processing of messages from other reporting nodes - Increased delay in data delivery to the fusion centre	[76], [77], [120], [152]–[155]

nine various EE methods may be adopted and may affect the values on the axes of the rose chart.

When analysing the two use cases presented above, the aforementioned rose charts and dependency matrices, one may have the following observations:

- the area enclosed by the solutions on the rose chart generally *relates* to the energy consumption of the considered use case. The higher the value on the rose chart axes, the higher the energy usage.
- However, this *relation* is not linearly proportional. The exact energy usage depends on specific parameters, like distances between nodes, which are not covered by the axes.
- Although energy consumption can be extrapolated from the rose chart, the energy efficiency cannot. The use case with a very limited energy consumption may *not* be energy efficient due to its low reliability or low throughput.
- There exist a high number of degrees of freedom in the CSS optimisation process. However, under specific use case constraints, the number of degrees of freedom may be significantly reduced. This may be observed in Fig. 14 where some values on the axes are strict and some are in a range. Therefore, instead of a polygon, one may observe a circled area which is explicitly the possible area of optimisation.

X. CONCLUSION

Cooperative sensing is an efficient method introducing additional detection gain at the cost of introduced communication overhead. In this survey we have briefly described the ways for obtaining energy-efficiency in cooperative sensing. The relevant methods can be applied in local (single-node) spectrum sensing procedures, in selection of cooperating, sensing, reporting and relaying nodes, in the application of the appropriate fusion rule, and finally in the proper network organisation. We briefly presented algorithms leading to relative energy saving while assuring high sensing performance in terms of the global probability of detection or the global probability of false alarm. These algorithms are classified according to main directions

of EE opportunities (called branches) in cooperative sensing. We also summarised (in Table V) the benefits and drawbacks of possible EE-increasing solutions, as well as recommended literature describing them in detail.

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